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منصة ذكية للتمريض عند الطلب وإدارة المرضى: Housepital-AI
**Housepital-AI: An Intelligent Platform for On-Demand
Nursing and Patient Management**

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ABSTRACT

The current healthcare landscape faces compounding challenges driven by demographic transitions and acute shortages of nursing professionals. Consequently, the traditional hospital-centric model of care delivery has become increasingly unproductive, cost-ineffective, and inaccessible for routine patient management. To bridge this critical gap, this project introduced Housepital-AI, an intelligent, cloud-native patient management platform designed to optimize real-time logistics for both home healthcare visits and in-clinic bookings. Developed over ten distinct agile sprints, the software-driven architecture was centrally managed through a modular, Role-Based Access Control (RBAC) dashboard, which enabled secure, role-specific administrative oversight for operations, quality assurance, and financial management. The system facilitated seamless end-to-end booking workflows supported by a geolocation-based routing engine equipped with dynamic pricing capabilities. To enhance user engagement and continuity of care, the platform integrated flexible subscription models that provided patients with a dynamic, personalized health dashboard. Furthermore, patient privacy and operational security were prioritized through a cloud-secured document verification module and a dual QR Code and PIN authentication system for rigorous visit verification. The logistical backend was augmented by advanced Python-based artificial intelligence microservices. This included a context-aware Natural Language Processing (NLP) chatbot designed for initial symptom classification, automated health education, and care reminders, alongside a dedicated Computer Vision module engineered for accurate wound analysis. Final system trials demonstrated exceptional robustness across all deployment scenarios. The Computer Vision microservice achieved a 92.3% accuracy rate in detecting infection symptoms, while the integrated emergency triage safety protocols recorded a 100% success rate. Ultimately, Housepital-AI establishes a highly scalable digital infrastructure that significantly improves workforce optimization, enhances patient-provider connectivity, and elevates clinical outcomes across diverse healthcare environments.

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LIST OF ACRONYMS/ABBREVIATIONS

ACRONYM	Full word/phrase of the acronym
ACID	Atomicity, Consistency, Isolation, Durability
AI	Artificial Intelligence
API	Application Programming Interface
AWS	Amazon Web Services
B2B	Business-to-Business
CI/CD	Continuous Integration / Continuous Deployment
CV	Computer Vision
EC2	Elastic Compute Cloud (AWS Service)
ECG	Electrocardiogram
EMR	Electronic Medical Record
ETA	Estimated Time of Arrival
FK	Foreign Key
HIS	Hospital Information System
IoT	Internet of Things
IV	Intravenous
KPI	Key Performance Indicator
NLP	Natural Language Processing

OLTP	Online Transaction Processing
OTP	One-Time Password
OWASP	Open Web Application Security Project
PDF	Portable Document Format
PIN	Personal Identification Number
POLP	Principle of Least Privilege
PRD	Product Requirements Document
PT	Physical Therapy
QA	Quality Assurance
QR	Quick Response (Code)
RBAC	Role-Based Access Control
RDBMS	Relational Database Management System
REST	Representational State Transfer
RPO	Recovery Point Objective
RTC	Real-Time Communication
RTO	Recovery Time Objective
S3	Simple Storage Service (AWS Service)
SHNSS	Smart Homecare Nursing Support Systems
SLA	Service Level Agreement
SMS	Short Message Service
SRE	Site Reliability Engineering

SUS	System Usability Scale
TLS	Transport Layer Security
UAT	User Acceptance Testing
UX	User Experience
WSN	Wireless Sensor Network
SP	Story Points

Chapter One

1. INTRODUCTION

1.1. Introduction

The current state of the global healthcare environment is also facing a paradigm shift with the ‘twin tsunamis’ of an ageing population and an increase in non-communicable or chronic diseases. Thus, with a forecast by the United Nations that sees tripled numbers of individuals aged over 80 by 2050, putting a tremendous burden on current healthcare systems [1], chronic non-communicable diseases also remain responsible for a massive 75% of current healthcare expenditure globally, calling for a shift from reactive therapy for acute illnesses towards a proactive program for chronic management at a domestic level [2].

Nonetheless, legacy hospital-based systems are still inefficient and costly. In the USA, MENA, and other regions in general, increased inflation levels coupled with huge growth in hospital expenditure have brought financial instability in health systems into effect [3], while the financial implication associated with hospital readmissions is persistently affecting health budgets globally in these regions [4]. Locally, within the Egyptian environment in relation to its digital revolution, there is a need to ensure alignment with ‘Egypt Vision 2030’ to promote health service digitization in achieving equal accessibility for all Egyptian citizens [5].

Housepital-AI responds to such challenges by providing a smart and on-demand nursing/patient management system. Housepital-AI takes advantage of the “gig economy” trend among nursing professionals, recognized through research to be a crucial paradigm for accommodating flexible work options and combating healthcare workforce rigidity [6]. Despite the expected exponential growth of the telehealth industry worldwide by 2030 [7], the “Uber for Nursing” phenomenon needs to be executed with great caution. It is evident from research findings that the unchecked growth of the “gig economy” trend may endanger healthcare quality unless embedded with simplified healthcare governance [8]. Housepital-AI

tackles such threats with its innovative blend of the “gig economy” trend with a stringent cloud-based verification process for protecting patient safety [9].

1.2. Motivation

The driving force behind the creation of Housepital is the utmost need to deal with the "twin tsunamis" that exist in the area of home care. Higher inflation and the remarkable rise in hospital costs are adding to these difficulties.

1. Inefficiency in Service Delivery and Accessibility for Patients: The traditional process for coordinating home care services is inefficient. It is even worse for patients requiring sudden services; they have to wait for a time that could quadruple during high demand. It is important to incorporate value-based healthcare approaches to tackle this issue. Healthcare should focus on patients' results rather than just concentrating on visiting numbers [10]. In addition, a study has discovered that relocating patients from hospitals to home environments not only reduces costs substantially but ensures a high standard of care is maintained too [11]. Housepital-AI proposes to ensure waiting times are less than 60 minutes.

2. Retention of Workers and Ensuring their Safety: The state of the healthcare workforce is presently facing a historic crisis. The current nurses in the healthcare system are experiencing a turnover rate of between 20% and 30% of the nurses in the system, mainly attributable to the “lone worker” phenomenon that occurs in the context of home healthcare, in addition to burnout. Current longitudinal studies have presently validated the relationship that does exist between the burnout of nurses and the adverse outcomes of patient safety [12]. Moreover, it is essential that the nurse staffing levels be sufficiently maintained in a positive manner in terms of influencing the outcomes of patient safety, in that shortages in staffing levels have been shown to be associated with an increase in mortality rates [13]. This current crisis in the healthcare workforce is, in turn, worsened by the current shortage of nurses globally, set to increase by 13 million in the year 2030 if the trends persist [14].

3. Healthcare Democratization using AI: There is still a major "digital divide" concerning health care. Rural and remote communities are affected by diagnostic delays due to the inability

to reach expert health facilities promptly [16]. Housepital-AI democratizes health care by enabling individuals who are not health experts to extract health-quality data, such as photos of a wound, for analysis by Deep Learning algorithms [17]. This satisfies the health strategy through digital health, as advised by the World Health Organization, to promote mobile health interventions for universal health coverage objectives [18]. Project Objectives

The prime target of the Housepital project is the development of an intelligent, on-demand nursing system that will connect the patients with accredited professionals, with the use of AI in decision support. The specific targets can be summarized as follows:

- **Intelligent On-Demand & Clinic Booking:** For implementing a dual-booking engine which helps in:
 - **Home Visits:** Automatic dispatching of nursing availability within 60 minutes via geospatial matching algorithm.
 - **Clinic Appointments:** An adaptable appointment scheduling solution for physicians incorporating both "Slot-based" and "Queue-based" appointment systems.
- **Family-Centric Care Management:** Managing elderly family members' care is an uphill task for families. Research indicates that family-centered digital interventions increase drug compliance and prevent hospitalizations by up to 25% in order to educate family caregivers [19]. Furthermore, the penetration of mobile technology in developing areas presents a great opportunity to offer these interventions at a massive level [20]. In addition, through the engagement of AI-powered triage Lite systems, it is possible to decrease avoidable hospitalizations in order to guide patients to receive care appropriate to their situations [21].
- **Service Delivery Security:** With the use of strict safety protocols through digital verification mechanisms, specifically using a PIN Code to confirm the arrival of home nurses and QR Codes for check-in purposes at clinics, the care will only be provided by authorized personnel.

- **AI-Powered Triage and Assistance:** To build an AI-driven chatbot that is context aware and performs "Triage Lite" in order to identify high-risk emergencies that need immediate notification and offers patient-specific health education resources based on the selected patient profile.
- **Computer Vision for Wound Assessment:** The application of the computer vision module for evaluating the images of low risk wounds to identify the presence of infection symptoms of redness and swelling, as an early warning system for post-operative patients.
- **Medical Supervision & Quality Control:** To create a specific panel for ‘Supervising Doctors’ and ‘Quality Specialists’ for the purpose of examining high-risk cases and authenticating credentials before being activated.”

Chapter Two

2. BACKGROUND AND LITERATURE REVIEW

2.1. Overview

The chapter above looks retrospectively into the literature related to AI in the healthcare industry. It outlines the theoretical background for the development of the Housepital-AI system based on the latest research related to intelligent systems and their application [22].

2.2. Artificial Intelligence and Chatbot Efficacy

Recent comparative studies have illustrated that AI is indeed proving to be effective in engaging with patients directly. Ayers et al. conducted a blinded assessment of 195 patient questions where 78.6% of the respondents preferred the AI response over the medical professional, citing higher levels of empathy and superior quality [23].

2.3. Computer Vision in Wound Assessment

Assessment of the physical conditions is a crucial component of tele-nursing. Superpixel segmentation was adopted in the pioneering study by Wang et al. to estimate the area of the wound. An accuracy of 94.2% was reported [24]. Such verification of the automatic assessment has laid the foundation of the Housepital-AI system for the remote categorization of the wound.

2.4. On-Demand Models and Patient Satisfaction

The "Uber-like" model has remained robust in the medical sector as well. In a case study on online medial services, it was observed that there was a staggering 168% rise in patient visits during health emergencies [25]. In spite of longer waiting times, patients remained extremely satisfied, with 98.3% being satisfied with communication options [26].

2.5. IoT and Smart Home Monitoring

For patients with chronic illness, continuous monitoring is essential. Smart Homecare Nursing Support Systems (SHNSS) with Wireless Sensor Networks have been effective in decreasing false calls to the ambulance with 20% through multi sensors integration [27].

2.6. Accessibility and Workforce Challenges

Barriers also exist within the systemic structure of home care. According to Ornstein et al., despite the efficiency of home care for dementia patients, important ethnic inequality also persists for home care services [28]. Moreover, staff turnover is a significant threat to home care, standing at 30% due to the loneliness experienced by visiting the home, making effective approaches to this essential to ensure continuity [29].

2.7. Data Security

New Models, incorporating Blockchain Technology, have been proposed for offering a ‘decentralized, transparent ledger’ for ensuring the integrity of medical records [30].

2.7. Comparative Analysis of Methodologies, Accuracy, and Complexity

Table 2-1 Comparative Analysis of Methodologies, Accuracy, and Complexity [16]

Ref.	Methodology	Metrics	Complexity / Scope	Advantages	Disadvantages
[22] Arai et al.	Theoretical Framework	Valid: Verified via functional architectural simulations.	High: System architecture design. & Automates routine education; reduces load.	Automates routine education; reduces load.	High implementation complexity.
[23] Ayers et al.	Cross-Sectional Blinded Study	78.6% preference for AI. AI quality: 4.13/5.	Medium: 195 queries; expert review.	High empathy and detail; scalable triage.	Risk of hallucination; lacks patient history.
[24] Wang et al.	Image Segmentation (Super pixel)	94.2% segmentation accuracy vs. ground truth.	High: Algorithmic development.	Objective, low-cost measurement.	Sensitive to lighting and image quality.
[25] Zettler-Greeley.	Retrospective Analysis	168% volume handling: Scale efficiency proved.	Medium: Single platform data (Jan-May 2020).	Proven elasticity and scalability.	Operational strain; increased wait times.
[26] Ramaswamy.	Cross-Sectional Survey	98.3% positive rating on communication.	Low: Patient survey.	High user acceptance; effective for consults.	Self-reported bias; excludes non-tech users.
[27] Lee et al.	Experimental Prototype	20% reduction in false emergency alarms	High: Hardware integration (WSN/ZigBee).	Continuous vitals tracking; auto-alerts.	Hardware maintenance; connectivity dependency.
[28] Ornstein et al.	Cohort Analysis (Medicare)	44.4% target reach (Dementia patients).	High: National dataset (n=6,664).	Targets vulnerable populations effectively.	High inequity (Hispanic pts 68% less likely).
[29] Drennan et al.	Scoping Review	Positive: Retention confirmed stable at 12 months	Medium: Synthesis of 7 qualitative studies.	Identifies key support needs (supervision).	"Lone worker" isolation remains a major risk.
[30] Siyal et al.	Blockchain Analysis	Valid: Cryptographic Integrity (Immutable Ledger).	High: Architectural proposal.	Immutable, tamper-proof patient records.	High computational overhead/storage cost.

Chapter Three

3. PROPOSED SYSTEM

3.1. Overview

Housepital-AI is an intelligent system aimed at connecting patients and nursing personnel using the logistics model. Housepital-AI is a two-way platform that performs the following roles:

- **On Demand Nursing:** The logistics motor that connects the patient with verified RNs in real time, like ride-sharing applications [31].
- **AI Health Assistant:** A dedicated module that utilizes NLP for medical guidance and Computer Vision to analyze low-risk wounds. Current research suggests that such integrated AI assistants can improve patient adherence to home-based recovery protocols by providing immediate, low-stakes feedback [32].

The system is aimed at catering to senior patients, those with chronic ailments, as well as post-operative patients, allowing for a completely digital experience as it encompasses appointment management, patient tracking, as well as patient care.

3.2. Why is this important?

The relevance of Housepital-AI is rooted in its capabilities to fix the essential gaps that exist within the current healthcare sector because of deficiencies in the management and governance of

- **Accessibility:** It tackles the problem of the “twin tsunamis” of restricted access and shortages of nursing staff through democratized access to nursing care [33].
- **Efficiency:** While legacy models are often bogged down by administrative delays, our system targets a service acceptance window of under 60 minutes. Utilizing geospatial optimization ensures that the closest qualified professional is notified immediately, reducing travel-related latencies [34].
- **Safety and Supervision:** Unlike standard telehealth, we include a "Supervision Interface". This helps doctors oversee high-risk cases and lessens the "lone worker" isolation that often leads to high nurse turnover—a strategy recently validated for improving long-term provider retention [35].
- **Cost Reduction:** Through the digitization of the booking and administrative process, the platform achieves efficiency by minimizing operational costs and offering transparent and dynamic pricing

3.3. Agile Framework Used

3.3.1. Definition of Scrum

Scrum is an iterative methodology used to manage complex, knowledge-based work. It prioritizes collaboration, self-management, and the flexibility to pivot when business realities shift. For Housepital-AI, this framework was essential due to the technical hurdles of blending Geospatial Tracking with Computer Vision. Implementing Scrum allows for the continuous refinement of security protocols, which is a mandatory standard in the evolution of safety-critical medical software [36].

In the case of Housepital-AI, the team used the Scrum framework because of the high level of uncertainty involved in integrating the technologies of Geospatial Tracking, Computer Vision, and Real-time Chat. With this, it was possible to produce “Potentially Shippable Product Increment” at the end of every sprint.

3.3.2. Reason for Choosing Scrum

The use of the Scrum methodology in the development of the Housepital AI was helped by some of the following critical factors in the development of the healthcare applications:=-

- **Complexity and Uncertainty Management:** By segmenting the project into sprints, we could isolate the development of AI models from the core logistics engine. Recent studies in software engineering show that this "modular sprint" approach is the most effective way to integrate deep learning into mobile platforms [37].
- **Early Risk Mitigation:** Safety comes first in medical apps. The Scrum process helped us catch hard failures early. Agile practices align with best practices for identifying, assessing, and mitigating risks in project management[17]. For instance, by testing the Triage Lite logic in Sprint 4, we were able to address safety issues before the end of the whole project, rather than at the very end [38].
- **Feedback and Capability of Adapting to It:** “Service Catalog” requirements kept changing during the analysis phase. Scrum helped us in adapting to the change in the database schema requirements in Sprint 3 without affecting the implementation of requirements in Sprint 1 [39].
- **Continuous Integration:** In order to avoid the "Integration Hell" situation common during the last few weeks of the graduation projects, by requiring a deployable build at the conclusion of every sprint cycle, "Patient App" and "Nurse App" were successively incorporated.

3.3.3. Scrum Roles

Table 3-1: Scrum Roles

Role	Responsibilities
Product Owner	Acted as the voice of the customer, prioritizing the backlog to ensure every feature delivered actual clinical value. This role is vital for bridging the gap between technical developers and medical requirements [40].
Scrum Master	Facilitates Scrum events, ensures adherence to Scrum principles, removes impediments, and supports continuous improvement within the team [40].
Development Team	Designs, implements, and tests product increments; delivers according to the Definition of Done. Self-organizes with the team to complete Sprint tasks [40].

3.4. Project Plan (Methodology and Phases)

Housepital-AI was brought to life over 4 Agile sprints:

- **Sprint 1: Foundation & User Ecosystem:** This phase focused on the core API and user management. We recognized that a single user must be able to manage multiple dependent profiles, which required a robust data relational model. Establishing a "Trust Framework" at this stage is a prerequisite for any modern telehealth system [41].
- **Sprint 2: Trust Layer & Admin Control:** Here, we built the verification tools for Quality Specialists to vet nurse credentials.
- **Sprint 3: Core Operations:** We developed the service catalog and the "fair pricing" algorithm. Our formula factors in base price, distance, time, and nurse experience. Modern logistics research indicates that such multi-variable pricing is essential for maintaining a healthy gig-economy ecosystem [42].
- **Sprint 4: AI & Computer Vision:** The final phase integrated high-risk keyword triggers and wound analysis models to enhance the platform's diagnostic capabilities.

Product Backlog

In scrum structure have inputs to start work, from inputs have product backlog, product backlog it is a list contain all product items, such as: user stories, bugs, technical work, features that is displayed in product backlog table . Also it is frequently updated/groomed according to sprints work and will define sprint in next part

Table 3-2: Product Backlog

ID	User Story	Acceptance Criteria	Est (SP)	Priority
OB-001	As a new customer I want to create an account and verify my identity So that I can securely use the platform	• User can enter personal information (name, email, phone number, password) • National ID photo capture is mandatory • Identity verification process is triggered automatically	8	High

		- Onboarding Wizard appears after successful registration - User can skip optional steps with clear indication		
OB-002	As a Customer I want to add myself as a patient in the app So that I can book services for myself	- User can enter basic medical information (date of birth, allergies, chronic conditions) - Confirmation message "Medical profile added successfully" is displayed - Medical profile is linked to the customer account - Profile is immediately available for service booking	3	High
OB-003	As a customer I want to add a minor dependent (child/elderly) So that I can manage their healthcare	- User can enter dependent's information (name, date of birth, relationship) - ID proof upload for dependent is mandatory - Status shows "Pending Review" until admin approval - User receives notification when dependent is approved/rejected - Approved dependents appear in patient selection list	5	High
OB-004	As a customer I want to invite an adult family member as a dependent So that they can consent to me managing their care	- User can send invitation via phone number or email - Invitee receives the invitation notification - Invitee can accept/decline the invitation - Invitee can set permission levels for the customer - Connection is established only after invitee's consent	5	Medium
OB-005	As an admin I want to review dependent verification requests So that I can ensure data accuracy and prevent fraud	- Pending verification requests appear in admin dashboard - Admin can view uploaded ID documents alongside entered data - Admin can compare and verify document authenticity - Admin can approve or reject with reason - Customer receives notification of the decision - Approved dependents are activated immediately	3	High
BK-001	As a customer I want to browse available services and select what I need So that I can book the appropriate healthcare service	- Services page displays all available services organized by categories - User can search and filter services easily - Each service shows name, description, and base price "Request Now" button is available for each service - Service details page shows full information before booking	5	High
BK-002	As a customer I want to choose which patient the service is for So that the correct person receives care	- Screen displays "Who is this service for?" prompt - List shows all registered patients (self + dependents) - Default patient is pre-selected (most recently used or self) - User can change selection before proceeding - Selected patient's medical info is passed to the booking	3	High
BK-003	As a customer I want to specify if I have medical supplies or need the nurse to bring them So that the service cost is calculated correctly	- System asks: "Do you have the required supplies or should the nurse bring them?" - If nurse brings supplies: deposit amount is displayed	5	High

		- User must pay deposit immediately via payment gateway - Payment confirmation is shown before proceeding - Deposit is clearly labeled as refundable (if applicable)		
BK-004	As a customer I want to schedule the service and add special requirements So that the service is delivered at my preferred time	- User can select: "Immediate" or "Scheduled" (calendar picker) - Prescription upload is available (mandatory for certain services) - Nurse gender preference option (Male/Female/Any) - Notes/special instructions field is available "Search for Nurse" or "Confirm Booking" button	5	High
BK-005	As a system I want to find the most suitable available nurse for the request So that customers get matched with qualified providers quickly	- System filters nurses by: specialty, location, gender preference - System ranks by: distance, rating, response speed - Request is sent to the most suitable nurse first - If rejected, request moves to next suitable nurse - Customer sees "Searching for nurse..." status	8	High
BK-006	As a nurse I want to receive and review service request details before accepting So that I can make an informed decision to accept	- Urgent push notification with sound for new request - Request card shows: service type, distance, estimated time - Earnings for this visit are clearly displayed - Medical supplies info is shown (customer has / nurse brings) - Customer rating is visible - Countdown timer for response (e.g., 60 seconds) - Customer rating is visible	5	High
BK-007	As a customer/nurse I want to communicate and track progress during an active visit So that both parties stay informed throughout the service	- Customer can track nurse location on live map - In-app chat is available between customer and nurse - In-app calling is available (without exposing phone numbers) - Visit can only start when nurse enters secure PIN - PIN verification confirms nurse arrival - Visit report can be filled by nurse during/after service "Complete Visit" button available after service delivery	8	High
BK-008	As a customer I want to receive the final invoice, pay, and rate the service *So that I can complete the transaction and provide feedback	- Final invoice shows: service cost + supplies cost - deposit paid - Multiple payment options: Cash / Card / E-wallet - Mutual rating system: customer rates nurse, nurse rates customer - Rating includes: star rating (1-5) + optional comment - Visit is closed automatically after ratings are submitted - Receipt is saved in booking history	5	High
CN-001	As a customer	Cancel button is available when booking is in "Pending" status	3	High

	I want to cancel my booking early (before nurse accepts/departs) So that I can get a full refund without penalties	• Cancellation before nurse acceptance triggers 100% deposit refund • Nurse is notified if already assigned but not departed • Booking status changes to "Cancelled - Customer (Early)" • Refund is processed automatically to original payment method • Cancellation reason is optional at this stage		
CN-002	As a system I want to apply late cancellation policy and compensate the nurse So that nurses are protected from last-minute cancellations	• Cancellation fee is deducted from customer • Customer loses the medical supplies deposit • Nurse compensation requires proof submission: - Photo of purchase receipt - Photo of opened/used medical supplies • Admin reviews and approves compensation request • Nurse receives compensation after admin approval • Booking status changes to "Cancelled - Customer (Late)"	8	High
CN-003	As a nurse I want to cancel an accepted visit for emergency reasons So that I can handle personal emergencies with minimal impact	• Nurse must select cancellation reason from predefined list • Customer receives apology notification immediately • System automatically re-queues the request with "Priority" flag • Request is sent to next available suitable nurse • Nurse's "Visit Completion Rate" metric is affected • Original nurse cannot accept the re-queued request • If no nurse found within timeout, customer is notified	5	Medium
CN-004	As a nurse I want to report when customer is not available at location So that I am compensated for my time and travel	• Nurse selects "Customer Not Available" option upon arrival • System starts 10-minute waiting timer • Nurse must prove contact attempt (in-app chat + call) • After timer expires, "Cancel - No Show" button activates • No-Show fee is charged to customer • Nurse receives compensation for travel/time • Booking status changes to "Cancelled - No Show" • Incident is logged for customer history	5	High
CN-005	As a nurse I want to report unexpected issues at the service location So that I can get support without negative impact on my rating	• Nurse can report issues: "Service Mismatch", "Safety Concern", "Other" • System escalates to priority support queue • Live support agent is connected for immediate assistance • Nurse rating is not negatively affected for reported issues • Admin investigates and resolves the situation • Customer and nurse are both notified of resolution • Visit is cancelled or rescheduled based on admin decision	3	Medium
DR-001	As a new doctor I want to register my account and verify my clinic	• Doctor can create basic account (name, email, phone, password) • Professional profile form includes: specialty, certifications, experience	8	High

	So that I can offer my services through the platform	- Clinic details form includes: name, address, photos, working hours - All required documents can be uploaded (license, certificates) - Profile is submitted for admin review and approval - Doctor receives notification when approved/rejected - Rejected applications show clear reason and resubmission option		
DR-002	As a doctor I want to define my services and assign them to specific clinics So that patients know what I offer at each location	- Doctor can add new service with: name, description, price, duration - Checklist shows all doctor's verified clinics - Doctor can select which clinics offer each service - Services can have different prices per clinic (optional) - Service visibility can be toggled on/off - Changes are reflected immediately in patient-facing app	5	High
DR-003	As a doctor I want to manage my availability and booking preferences So that I have control over my schedule and workflow	- Booking method selection: Time Slots vs. Queue System - Time Slots: Define available days and specific time slots - Queue System: Set daily patient limit and estimated time per patient - Minimum advance booking time setting (e.g., 3 hours before) - Urgent/Same-day appointments toggle with premium pricing option - Buffer time between appointments setting - Vacation/unavailable dates blocking	8	High
DR-004	As a patient I want to search for doctors and book a clinic appointment So that I can receive medical consultation at a clinic	- Search by: specialty, location, doctor name, availability - System auto-hides appointments within minimum booking time "Urgent" filter shows same-day available slots (if enabled by doctor) - Doctor profile page shows: bio, photo, rating, services, clinics [] Available slots displayed in calendar or list view [] Patient can select preferred clinic if doctor has multiple [] Clear display of consultation fee before booking	8	High
DR-005	As a patient I want to pay for my appointment upfront and get confirmation So that my slot is secured and I have all booking details	- Payment page shows: service, doctor, clinic, date/time, amount - Multiple payment methods: Card, E-wallet, Saved payment methods - Payment is processed before booking is confirmed - Confirmation notification sent via Push + SMS - Booking appears in "My Appointments" section - Appointment details include: address, directions, preparation notes - Calendar event can be added to phone calendar	5	High
DR-006	As a system I want to enforce cancellation policies for clinic bookings	- Early cancellation (>24 hours): 100% refund - Late cancellation (<24 hours): 75% refund - Very late cancellation (<3 hours): 50% refund - No-show: 0% refund	5	High

	So that doctors and patients are treated fairly	• Doctor cancellation: 100% refund + bonus credit for patient • Doctor cancellation affects "Reliability Score" • Automatic refund processing to original payment method • Both parties notified of cancellation with reason		
DR-007	As a doctor/patient I want to verify patient arrival using QR or PIN So that attendance is confirmed and consultation can begin	• Doctor's app displays unique QR Code for each appointment • 4-digit PIN is shown alongside QR Code • Patient presses "Check In" when arriving at clinic • Patient can scan QR Code using phone camera • Alternative: Patient can enter PIN manually • Successful verification changes status to "Checked-In" • Doctor is notified when patient checks in • Check-in timestamp is recorded • Protection against false "I arrived" claims	5	High
NR-001	As a new nurse I want to register and set up my professional profile So that I can start offering my services on the platform	• Nurse can create basic account (name, email, phone, password) • Mandatory document uploads: - National ID card - Nursing degree certificate (college or accredited institute) - Valid nursing license from the Syndicate • Professional profile setup: - Specialties and skills (wound care, IV, injections, elderly care, etc.) - Years of experience - Bio/About me section (visible to customers) • Bank account or e-wallet linking for payments • Profile submitted for admin review • Status shows "Pending Verification"	8	High
NR-002	As an admin I want to review and verify nurse applications So that only qualified nurses can offer services	• Pending nurse applications appear in admin dashboard queue • Admin can view each uploaded document clearly • Admin can verify document authenticity and validity • Optional: Video call verification with nurse • Approval: Nurse receives welcome notification, can start working • Rejection: Clear reason sent, nurse can resubmit corrected docs • Verification status is tracked with timestamp	5	High
NR-003	As a nurse I want to control when and where I'm available for work So that I only receive relevant requests	• Toggle switch on home screen: Online / Offline • Online: Nurse appears in matching system, receives requests • Offline: Nurse is hidden, no requests sent • Work zone setup on map: - Set center point (home or preferred location) - Set service radius (e.g., 5km, 10km, 15km) • Only requests within zone are sent to nurse • Schedule availability: Set working days/hours (optional) • Auto-offline after certain idle time (configurable)	5	High
NR-004	As a nurse I want to receive clear request details before accepting So that I can make informed decisions about each job	• Urgent push notification with distinctive sound • Request card displays: - Service type (e.g., "Wound Care and Dressing Change") - Distance and estimated travel time - Expected earnings for this visit - Medical supplies info (customer has / nurse brings) - Customer rating and number of completed bookings • Countdown timer for response (e.g., 60 seconds)	5	High

		- Accept button: Nurse is assigned, customer notified - Reject button: No penalty, but affects priority if frequent - Request expires if no response within timeout		
NR-005	As a nurse I want to track my earnings and withdraw money So that I can manage my income from the platform	- Dedicated "My Earnings" page in nurse app - Display sections: - Available Balance (ready for withdrawal)- Pending Balance (not yet settled) - Total Earnings (all time) - Transaction history with details: - Each visit earning - Platform fees deducted - Bonuses and - Cancellation compensation - Withdraw button with amount input - Confirmation before processing withdrawal - Transfer to linked bank/e-wallet within 2-3 business days - Minimum withdrawal amount enforced	8	High
NR-006	As a nurse I want to view my visit history and performance statistics So that I can track my progress and improve	"My Visits" page with list of all past visits - Filter by: date range, service type, status - Each visit shows: date, service, customer, earnings, rating - Performance dashboard: - Total visits - Average - Completion rate - Response time average - Earnings this week/month - Rating breakdown (5-star, 4-star, etc.) - Customer comments/feedback visible to nurse	5	Medium
NR-007	As a nurse I want to receive timely push notifications So that I never miss important updates	- Notification types: - New service request (high priority, with sound) - Request accepted - Customer message received - Visit reminder (30 min before scheduled) - Payment received - Rating received - Account/verification updates - Notification settings page to customize preferences - Silent hours option (e.g., 11 PM - 7 AM) - Critical notifications override silent mode	3	High
NR-008	As a nurse I want to manage my linked payment accounts So that I can receive my earnings conveniently	- Add bank account (account number, bank name, holder name) - Add e-wallet (Vodafone Cash, Fawry, etc.) - Set default payment method for withdrawals - Edit existing payment methods - Delete payment method (if not default) - Verification step for new payment methods - At least one payment method required to start working	5	High
AI-001	As a user I want to understand the chatbot's capabilities and limitations So that I use it appropriately and seek emergency help when needed	- Clear "AI Assistant" icon/button visible in the app - First interaction shows mandatory welcome message: - What the assistant can - What it CANNOT do (not a doctor, not for emergencies) - Emergency instructions (call 123, go to nearest hospital) - Emergency symptoms listed: chest pain, difficulty breathing, etc. - User must tap "I Understand & Agree" to proceed - Disclaimer is shown once per session or after 24 hours	3	High
AI-002	As a user I want to chat about a specific patient's health So that the AI can give relevant personalized responses	- Before starting chat, prompt: "Who is this conversation about?" - List of registered patients (self + dependents) displayed - User selects patient (e.g., "My mother - Fatma")	5	Medium

		- Chat opens with header showing: "Discussing: [Patient Name]" - AI has access to patient's medical history for context: - Age, chronic conditions, allergies - Recent visits and diagnoses - User can switch patient mid-conversation - Context is maintained throughout the session		
AI-003	As a user I want to describe symptoms and get an initial assessment So that I know whether to seek immediate care or book an appointment	- User can describe symptoms in natural language - AI classifies severity into three levels: - LEVEL 1 - MILD: - Response: General advice and home care tips - Example: "I have a mild headache" - LEVEL 2 - MODERATE - Response: Recommend booking a doctor appointment - Quick action button to book - Example: "My child has had fever for 2 days with rash" - LEVEL 3 - EMERGENCY: - Response: Urgent red alert message - Display emergency number: 123 - One-tap call - Example: "I have severe chest pain and shortness of breath" - AI asks follow-up questions to refine assessment - Clear disclaimer that this is not a medical diagnosis - Option to "Book Service" or "Talk to Doctor" after assessment	8	High
AI-004	As a user I want to ask general health questions and get educational information So that I can learn about health topics	- User can ask general health questions: - "What are symptoms of diabetes?" - "How to prevent high blood pressure?" - "What foods are good for heart health?" - AI provides accurate, easy-to-understand responses - Responses include disclaimers when appropriate - Links to trusted sources (optional) - Suggested related topics for further reading - Content is localized (Arabic/English)	5	Medium
AI-005	As a user I want to set medication reminders through the chatbot So that I don't forget to take my medicines	- User can say: "Remind me to take blood pressure medicine at 8 AM" - AI parses the request and confirms - Medicine - Time(s) - Frequency (daily, weekly, etc.) - User confirms the correct details - Reminder is being created and saved - Push notification at the specified time - Snooze option on reminder notification - User can manage reminders via chat ("Show my reminders", "Delete reminder")	5	Medium
AI-006	As a user I want to escalate to human support when AI can't help So that I can get assistance for complex issues	- User can say "Talk to support" or "I need human help" - AI confirms escalation: "Connecting you to our support team..." - Seamless handoff to live support chat - Conversation history is passed to support agent - If support unavailable, offer callback option - Estimated wait time displayed	5	Medium
PL-001	As an admin I want to create and manage subscription plans and packages So that customers have flexible purchasing options	- SUBSCRIPTION (Recurring): - Setting the price for: Monthly / Quarterly / Yearly - Define included features/benefits - Set renewal rules (auto-renew, manual - Discount for longer terms - PACKAGE (One-time): - Setting the fixed price (one-time payment) - Define included services	5	Medium

		(e.g., "5 nurse visits") - Setting the ultimate expiry period (e.g., valid for 6 months) - No auto-renewal • Enable/disable plans without deleting • Set target audience (all users, new users, etc.) • Preview how plan appears to customers • Track active subscribers per plan		
PL-002	As a customer I want to browse available plans and subscribe to one So that I can get better value for healthcare services	• "Plans & Offers" section visible in app menu/home • Clear comparison of all available plans - Plan name and description - Price (per month/year or one-time) - Features - Savings compared to pay-per-use • "Subscribe Now" button for each plan • Payment page with selected plan details • Multiple payment options (Card, E-wallet) • Confirmation with receipt • Immediate activation after payment • Welcome notification with plan details	5	Medium
PL-003	As a subscriber I want to use my plan benefits when booking services So that I get the discounts/free services I paid for	• During booking, system checks for active subscription • If plan covers the service: - Show "Covered by your plan" badge - Apply discount or mark as free (based on plan) • If plan has service limits (e.g., 5 visits): - Show remaining count: "3 of 5 visits remaining" - Warn when approaching limit • If service not covered: - Show regular price - Offer: "Upgrade your plan to include this" • Plan benefits tracked and updated in real-time	5	Medium
PL-004	As a premium subscriber I want access to exclusive features and personalized content So that I feel the value of my subscription	• Home screen transforms to "My Health Dashboard": - Personalized greeting - Quick access to subscribed - Health tips tailored to patient profiles • Dynamic content sections: - "Today's Health Tip" based on conditions - "Your Care Reminder" for upcoming appointments - "Activity Tracker" integration (optional) - Recommended articles based on history • Premium badge visible on profile • Priority support queue access • Exclusive discounts on non-covered services • "Personal Health Coach" feeling throughout app	8	Low
PL-005	As a subscriber I want to view and manage my subscription So that I have control over my plan	• "My Subscription" page showing: - Current plan name and status - Start date and renewal date - Price and billing cycle - Features included - Usage stats (visits used, etc.) • Actions available: - View payment history - Update payment method - Upgrade to higher plan - Downgrade plan (effective next cycle) - Cancel subscription (with confirmation) - Enable/disable auto-renewal • Cancellation shows what will be lost • Grace period after expiry before deactivation	5	Medium
PL-006	As a subscriber I want to receive important subscription notifications via SMS/Email So that I never miss billing or expiry information	• Notification triggers:- Subscription activated- Payment successful (with receipt) - Renewal reminder (7 days before - Renewal reminder (1 day before) - Renewal successful - Payment failed (retry instructions) - Subscription expiring (if no auto-renew) - Subscription cancelled	3	High

		- User can choose channels: SMS, Email, or both - Unsubscribe link in emails - Notifications include plan details and action links		
AD-001	As an owner/super admin I want to manage my team and assign roles So that each employee has appropriate access levels	"Team Management" section in admin dashboard - Add new team member - Enter employee details (name, email, phone) - Select role from predefined - Set permissions (if custom roles allowed) - Employee receives invitation email to activate account - Edit existing team member's role/permissions - Deactivate/reactivate team members - View team member activity history - Only Owner can access this section	8	High
AD-002	As a quality & vetting specialist I want to review applications from doctors and nurses So that only qualified providers join the platform	"Pending Verification" queue shows new applications - Application details view - Personal information- Uploaded documents (ID, license, certificates) - Professional details - Document viewer with zoom capabilities - Verification checklist for each document type - Actions: Approve / Request Changes / Reject - Rejection requires reason selection/input - Approval triggers welcome notification to provider - Request Changes allows specific feedback - All actions logged with timestamp	5	High
AD-003	As a quality & vetting specialist I want to monitor provider performance and take action So that service quality is maintained	- System auto-flags providers with: - Low ratings (<4.0 average)- High cancellation rates (>10%) - Customer complaints - Flagged providers list with severity indicators - Provider detail view showing - Rating history and - Customer feedback/comments - Cancellation history with reasons - Response time metrics - Actions available - Send warning notification - Schedule quality review call - Temporary suspension - Permanent deactivation - Action history tracked for each provider	5	Medium
AD-004	As an operations manager I want to see live operations on a map So that I can intervene when issues arise	- Live map showing: - Active bookings (color-coded by status)- Available nurses (green dots) - Customers waiting (orange markers) - Filter by: service type, area, time - Click on booking to see details - Service type- Nurse assigned (or pending) - Time elapsed - Customer info - Interventions possible: - Expand search radius for unassigned - Manually assign nurse - Contact nurse - Contact customer - Alerts for delayed bookings (configurable threshold) - Real-time updates (WebSocket)	8	High
AD-005	As a customer support specialist I want to receive and resolve customer complaints So that issues are resolved quickly and fairly	- Complaints queue with auto-categorization: - Financial (refunds, overcharges - Service quality - Provider - App issues - Other - Complaint details view: - Customer info and history - Related booking details - Visit report (if applicable) - Chat history (if applicable)	5	High

		- Investigation tools - Contact customer (call/chat) - Contact provider (call/chat) - View location history - Resolution actions: - Issue refund (within limits) - Apply credit - Send apology - Escalate to manager- Flag provider for review - Resolution tracking with SLA timers		
AD-006	As a finance officer I want to manage platform finances and approve payments So that financial operations run smoothly	- Financial dashboard showing: - Today's revenue - Pending payouts - Refund requests- Outstanding issues - Compensation approvals: - Review nurse compensation claims - View attached proof (receipts, photos - Approve or reject with reason - Provider payouts - Scheduled payout list - Manual payout trigger (if needed - Payout history - Refund management: - Pending refunds queue - Process refund - Track refund status - Financial reports:- Daily/weekly/monthly revenue- Export to Excel/PDF - Send to management	5	High
AD-007	As an operations manager I want to use AI predictions to optimize operations So that we meet demand efficiently	"Demand Forecast" dashboard showing: - Predicted hot zones (heatmap) - Expected demand by hour - Suggested nurse deployment - Historical vs actual demand comparison "Dynamic Incentives" tool:- Create bonus for specific zone/time - Set bonus amount- Target specific nurse segments - Track incentive effectiveness - Alerts for predicted demand spikes - Integration with nurse notification system	8	Low
AD-008	As an owner I want to review all admin actions in audit logs So that I maintain accountability and security	- Audit log page (Owner only) - Every admin action logged - Who (employee name and role) - What (action description) - When (timestamp) - Details (affected records) - Example: "2025-01-14, 10:15 AM: [Support - Ahmed] refunded 50 EGP to [Customer - Mohamed Ali]" - Search and filter: - By employee- By action type - By date - By affected entity - Export audit logs (for compliance) - Log retention period configurable	5	High
AD-009	As an admin team member I want a dashboard customized for my role So that I see only relevant information	- Dashboard composed of modular widgets - Each role has predefined widget set: - Operations Manager: - Live map widget - Pending requests count - Active visits - Alerts widget - Quality Specialist:- Pending verifications count - Flagged providers widget - Recent reviews widget - Support Specialist - Open tickets - Priority tickets widget - SLA status widget - Finance Officer - Today's revenue widget - Pending payouts widget - Refunds queue widget - Clean, focused interface per role - Optional: Allow widget rearrangement	5	Medium
AD-010	As an admin user I want to recover my password if forgotten So that I can regain access to my account	"Forgot Password" link on admin login page - Enter email address - Receive password reset email with secure link - Link expires after 1 hour	3	High

		- Set new password (with strength requirements) - Confirmation and redirect to login - Notify owner of password reset (security measure) - Lock account after 5 failed attempts		
AD-011	As an admin user I want to switch the dashboard language So that I can work in my preferred language	- Language selector in settings/header - Available languages: Arabic, English - Immediate interface language change - Preference saved per user - All UI elements translated - Date/number formats adapt to locale	3	Medium
AD-012	As an admin I want to manage platform content without code changes So that we can update services and information easily	- Content Management sections: - Services Management - Add/edit/delete service categories- Add/edit/delete - Set base prices - Upload service images - Enable/disable services - FAQ Management - Add/edit/delete FAQ items - Organize by category - Set display order - App Content - Edit terms and conditions - Edit privacy policy - Edit about us - Banners & Promotions: - Upload home screen banners - Set banner links - Schedule banner display - Preview changes before publishing - Version history for content changes	8	MedZium
UI-001	As a user, I want to switch the app to Arabic so that I can navigate using my native language.	- Language toggle is available in settings. - UI fully flips to Right-to-Left (RTL). - All static strings are translated accurately.	5	High
UI-002	As a user, I want to enable Dark Mode so that the app is easier to read in low light.	- Theme toggle is available in settings. - High contrast dark palette is applied system-wide. - Icons and illustrations invert/adjust properly.	5	Medium
ALG-001	As a system, I want to execute a geospatial search within a defined radius to find online nurses.	- System queries PostgreSQL for nurses within 5km. - Query updates live based on nurse GPS coordinates. - Results are returned in under 500ms.	8	High
ALG-002	As a system, I want to calculate dynamic ETA based on traffic and distance for the matched nurse.	- Integration with Maps API for real-time routing. - ETA is calculated and displayed to the customer. - Nurse app shows optimized route.	8	High
ALG-003	As a system, I want to re-route a request immediately if a nurse rejects the match or times out.	- System maintains an active queue of eligible nurses. - Automatically pings the next nurse within 5 seconds of rejection/timeout. - Logging of rejection reasons.	8	High
ALG-004	As an admin, I want to adjust the matching algorithm's weights (e.g., rating vs. distance) dynamically.	- Admin dashboard contains a "Matching Weights" slider. - Changes update the algorithm logic without deployment. - Logs record who made the algorithm change.	5	Medium
BK-009	As a customer, I want to track the nurse's live location on a map after they accept my booking.	- WebSocket connection established for live tracking. - Map UI shows moving nurse icon. - Auto-updates ETA based on movement.	8	High
PAY-001	As a customer, I want to pay for my service using a credit/debit card securely.	- Integration with PayMob/Stripe API. 3D Secure verification supported. - Booking status updates to "Paid" via webhook.	8	High

PAY-002	As a customer, I want to pay using local e-wallets (e.g., Vodafone Cash) for accessibility.	• Mobile wallet payment option in checkout. • OTP/Phone verification for wallet deduction. • Instant receipt generation.	5	High
PAY-003	As a nurse, I want to view my digital wallet balance and request payouts.	• Wallet UI displays Current Balance and Pending. • "Request Payout" button available. • Deduction of platform commission is transparent.	5	Medium

3.5. Business Model

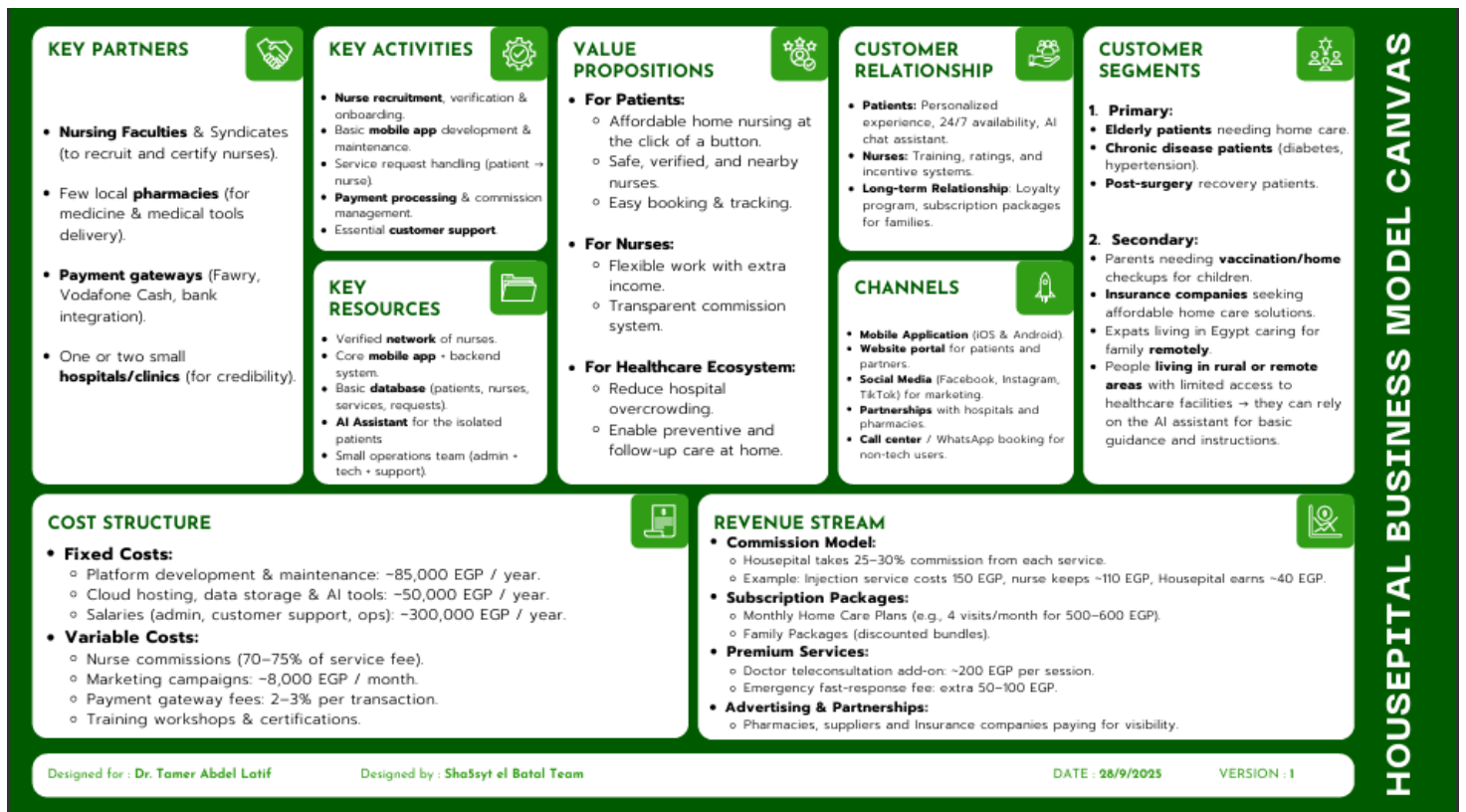


Figure 3-1: Housepital-AI Business

Sprint 1: Foundation & User Ecosystem

- **Goal:** the Backend API infrastructure with the ability to manage users.
- **Date:** [25/9/25] – [02/10/25]
- **Execution Phases:**
 1. **Analysis:** Analyzed the "Care Manager" section, realizing the necessity of having a single person manage multiple dependent profiles (e.g., parents and kids) along with different medical histories for each.

Table 3-3: User Stories for Sprint 1

ID	User Story (Backlog)	Task	Estimate	Status
OB-001	As a new customer I want to create an account and verify my identity So that I can securely use the platform	- Create User model with all fields - Implement register endpoint - Add email/phone validation - Implement OTP	2h + 2h + 1h + 2h	Done
OB-002	As a Customer I want to add myself as a patient in the app So that I can book services for myself	- Add medical info fields to User model - Create updateMedicalInfo endpoint - Create getMedicalInfo endpoint - Create medical info form UI - Form validation and submission	1h + 2h + 1h + 2h + 1h	Done
OB-003	As a customer I want to add a minor dependent (child/elderly) So that I can manage their healthcare	- Create Dependent model - Create addDependent endpoint - Create getMyDependents endpoint - Create updateDependent endpoint	1h + 2h + 1h + 1h	Done
AUTH-001	As a user I want to login to my account securely So that I can access my healthcare dashboard	- Implement login endpoint - JWT token generation - Auth middleware for protected routes	2h + 1h + 2h	Done
AUTH-002	As a user I want to reset my password if I forget it So that I can regain access to my account	- Forgot password endpoint - Reset token generation & email - Reset password endpoint - Forgot password UI - Reset password UI	1h + 2h + 1h + 1h + 1h	Done

2. **Design:** Worked on the PostgreSQL database designs for the Users, Family Members, and Addresses tables. API designs completed for Authentication and Profile Management.

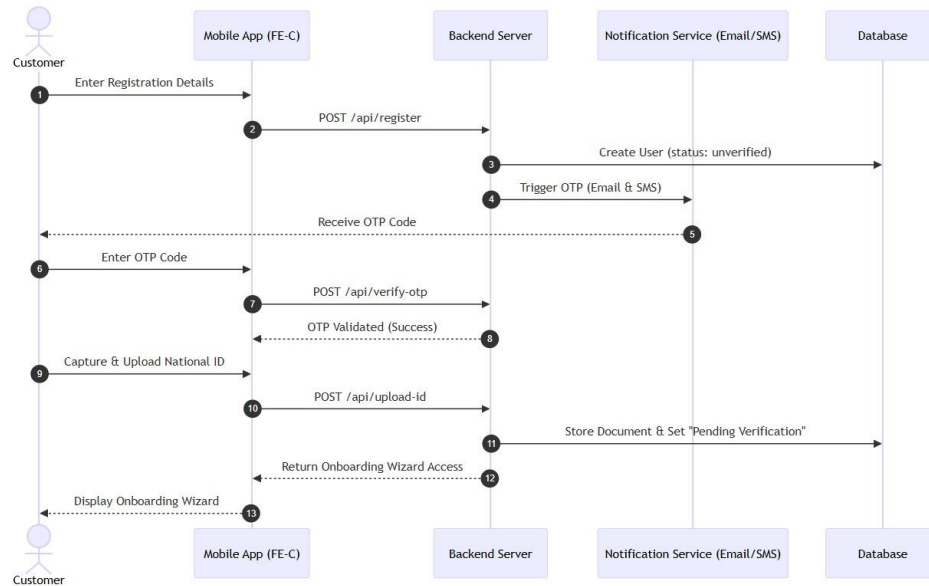


Figure 3-2: Sprint 1 Sequence Diagram: Customer Registration & ID

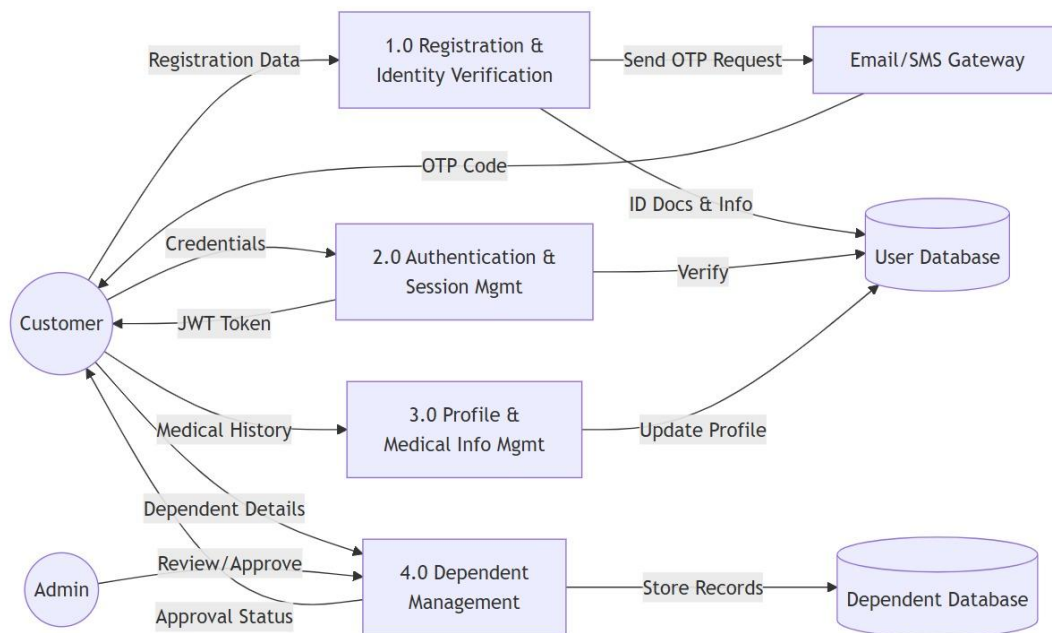


Figure 3-3: Sprint 1: Data Flow Diagram (DFD Level 1): Authentication & Onboarding

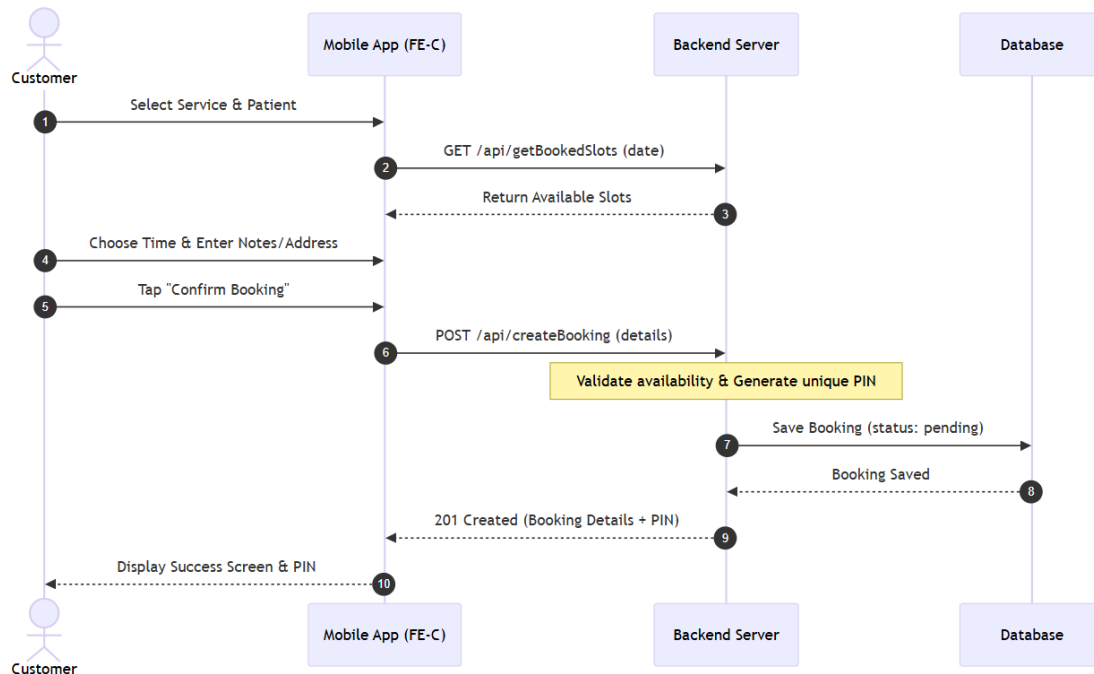


Figure 3-4: Sprint 1 Adding a Minor Dependent

3. Implementation:

- **Authentication:** Developed secure Login/Register APIs using JWT and bcrypt.
- **Profile & Settings:** established endpoints to manage user personal data and app configuration.
- **Family & Medical Info:** Implemented logic to store chronic conditions and dependent profiles.
- **Saved Addresses:** finished geospatial logic to save multiple delivery locations (Home, Parents' Home).

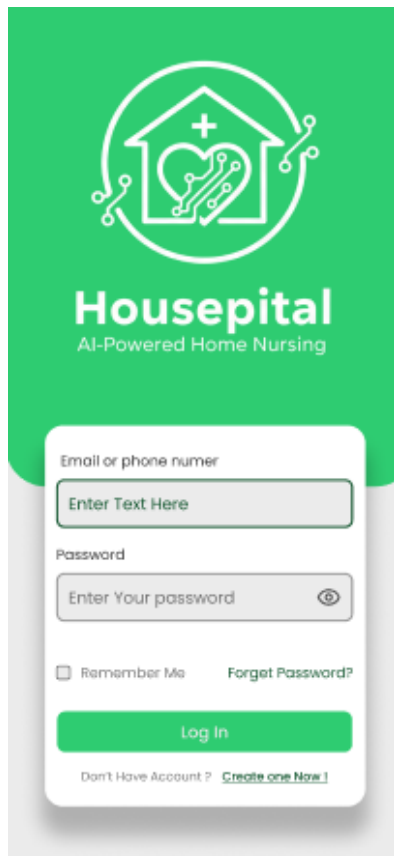


Figure 3-5: Login

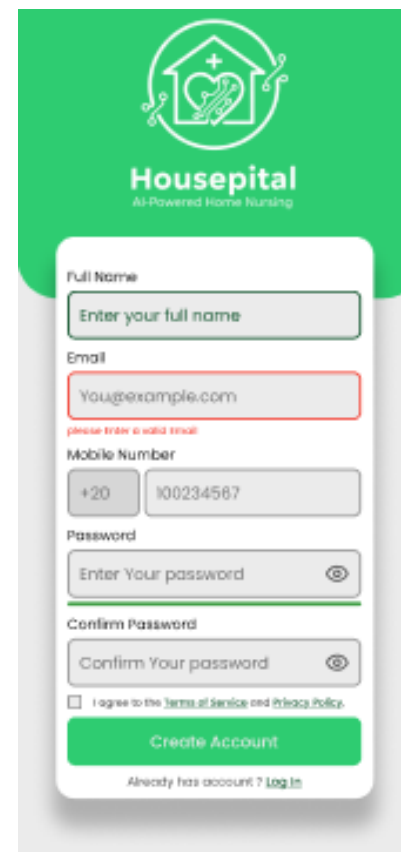


Figure 3-6: Register

4. Testing:

- did Unit Tests on Auth tokens to ensure session security.
- Tested "Address Saving" to verify GPS coordinates are stored the right way.

5. Deployment:

- Deployed the Node.js Core API and PostgreSQL Database to the AWS Development Environment.

Sprint 1 Review:

Incomplete Items - Back to Backlog:

- **Accomplishments:** Successfully made established the core important database architecture and finalized the user authentication workflow, including OTP validation.
- **Key Technical Deliverables:**
 - **Schema:** It is implemented the relational tables for Users, Family Members, and Saved Addresses.

- **Security Layer:** We deployed the bcrypt password hashing and JWT (JSON Web Token) session management.
- **Verification Logic:** After that, finalize the multi-channel OTP (One-Time Password) service for identity validation.

Sprint #1 Retrospective:

- **What worked well?** The relational data model for accommodating multiple dependents in the same account performed very well for the “Care Manager” use case.
- **What Could Be Improved?** It took longer than anticipated to set up the cloud-based development environment because of the intricacies involved in configuring it.
- **What Action Plan for Next Sprint?** Use standardized Docker containers to ensure equality between personal development environments and the development environment on the AWS server.

Sprint 2: Trust Layer & Admin Control

- **Goal:** Implement security verification via Cloudinary and build administrative tools.
- **Date:** [02/10/25] – [16/10/25]
- **Execution Phases:**
 1. **Analysis:** Defined the "Nurse Onboarding" workflow requiring manual verification of documents by Quality Specialists. The vetting of the process was trying to be designed to follow a step-by-step credentialing guide for providers[18] and is trying to adhere to the Nurse Licensure Compact standards[19].

Table 3-4: User Stories for Sprint 2

ID	User Story (Backlog)	Task	Estimate	Status
BK-001	As a customer I want to browse available services and select what I need So that I can book the appropriate healthcare service	- Create Service model - Create getServices endpoint (public) - Create getServiceById endpoint	1h + 2h + 1h	Done
BK-002	As a customer I want to choose which patient the service is for So that the correct person receives care	- Create patient selection component - Fetch user + dependents list - Default patient selection logic	2h + 1h + 1h	Done
BK-004	As a customer I want to schedule the service and add special requirements So that the service is delivered at my preferred time	- Add scheduling fields to Booking model - Implement getBookedSlots endpoint - Create date/time picker component	1h + 2h + 2h	Done
BK-005	As a system I want to find the most suitable available nurse for the request So that customers get matched with qualified providers quickly	- Create Booking model with full schema - Create createBooking endpoint - Generate booking PIN	2h + 3h + 1h	Done

BK-006	As a nurse I want to receive and review service request details before accepting So that I can make an informed decision to accept	- Create <code>getMyBookings</code> endpoint - Create <code>getBookingById</code> endpoint - Bookings list page UI - Booking detail page UI - Status badges and indicators - Pull-to-refresh functionality	2h + 1h + 3h + 2h + 1h + 30m	Done
CN-001	As a customer I want to cancel my booking early (before nurse accepts/departs) So that I can get a full refund without penalties	- Implement <code>cancelBooking</code> endpoint - Add cancellation reason field - Cancel button in booking details - Cancellation confirmation dialog - Update booking status in UI	2h + 30m + 1h + 1h + 30m	Done

2. Design:

- Designing the ultimate **Web Admin Panel** UI for managing the users.
- Designing the External Admin Dashboard for specifically for the "Verification Queue".

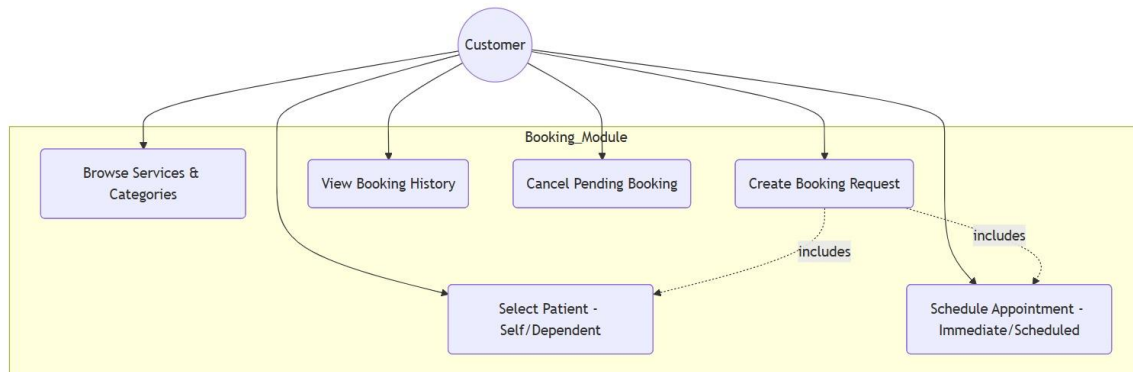


Figure 3-7: Use Case Diagram: Customer Booking

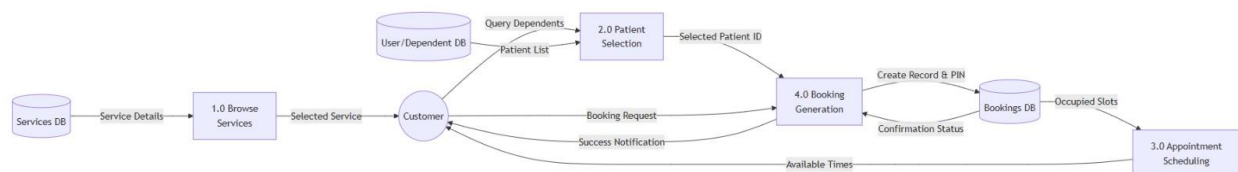
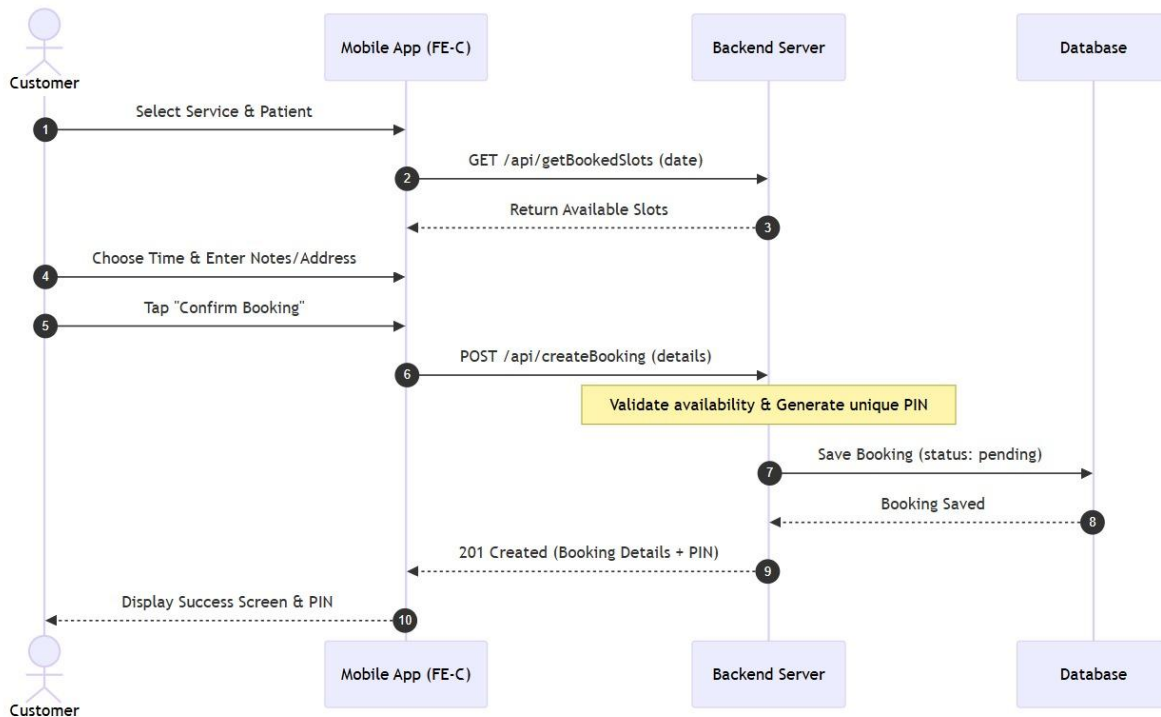


Figure 3-9: Data Flow Diagram (DFD Level 1): Booking

Figure 3-8: Sequence Diagram: Create Booking Request

3. Implementation:

- **Identity Verification:** Implemented the Cloudinary SDK for trying to make secure upload and the optimization of National ID cards and Nursing Licenses.



- **Admin Panels:** Created Web Admin Panels using React to help, as well as a specialized External Dashboard to check credentials for secure.

4. Testing:

- **Integration testing:** Verified that an image could be uploaded from the Mobile App appears correctly in the Admin Dashboard via Cloudinary.

5. Deployment:

- Deployed the Admin Web Client to the best AWS S3 (Static Hosting) and updated the Backend with verification logic code.

Sprint #2 Review:

- **Accomplishments:** Integrated secure cloud storage for professional documents and built the administrative verification tools.
- **Key deliverables:**
 - **Cloudinary Integration:** Developed the SDK integration for upload and auto-thumbnails generation for National IDs and Nursing Licenses.

- **Admin Control Panel:** Created a web dashboard specific to the manual review and verification of credentials of healthcare providers.
- **Trust Framework:** Implemented the logic for the 'Pending Verification' status so that the platform accesses are limited until the document gets approved.

Sprint #2 Retrospective:

- **What went well:** The asynchronous document upload feature led to improvements in the response time of the nurse registration process.
- **What Could Be Improved:** Handling state management within the multi-step "Onboarding Wizard" became challenging for tracking on various mobile screens.
- **What Action Plan for Next Sprint:** Transition to a more robust state management pattern in Flutter to ensure UI consistency during complex data entries.

Sprint 3: Core Operations & Booking Engine

Goal: Enable the core business Catalog connecting patients to the best of Home Care services.

- **Date:** [16/10/25] – [12/11/25]
- **Execution Phases:**
 1. **Analysis:**
 - Analyzed the Service Catalog requirements (Pricing, Duration, Risk Level).
 - The clinic scheduling logic deals with the issue of medical appointment scheduling by making use of ASP solutions and conventional clinic booking systems [20].
 - Dynamic Pricing Logic:
 - The requirements for a fair pricing algorithm were identified. A fair pricing algorithm was designed, where the final service price (P) is computed through a multivariable formula incorporating service base price (B), travel distance (D), time of day (T), and nurse-level experience (L)

$$P = B (\text{service}) + D (\text{Distance}) + T (\text{Time}) + L (\text{Level})$$

Table 3-5: User Stories for Sprint 3

ID	User Story (Backlog)	Task	Estimate	Status
DR-001	As a new doctor I want to register my account and verify my clinic So that I can offer my services through the platform	- Create Doctor model with all fields - Create doctor profile endpoint - Document upload for license/certificates - Staff app - Doctor registration flow	2h + 2h + 1h + 2h	Done
DR-002	As a doctor I want to define my services and assign them to specific clinics So that patients know what I offer at each location	- Create Clinic model - Create addClinic endpoint - Create getMyClinic endpoint - Create updateClinic endpoint	2h + 2h + 1h + 2h	Done

DR-002	As a doctor I want to define my services and assign them to specific clinics So that patients know what I offer at each location	- Link Service model to Doctor - Create addService endpoint (doctor) - Create getMyServices endpoint - Service-Clinic association	1h + 1h + 1h + 1h	Done
DR-004	As a patient I want to search for doctors and book a clinic appointment So that I can receive medical consultation at a clinic	- Create getAllDoctors endpoint (public) - Create getDoctorDetails endpoint - Search filters (specialty, name, area)	2h + 2h + 2h	Done
DR-005	As a patient I want to pay for my appointment upfront and get confirmation So that my slot is secured and I have all booking details	- Create getDoctorAppointments endpoint - Add booking settings to Doctor model - Create updateBookingStatus endpoint	2h + 1h + 2h	Done
DR-007	As a doctor/patient I want to verify patient arrival using QR or PIN So that attendance is confirmed and consultation can begin	- Add PIN field to Booking model - Generate PIN on booking confirmation - Create checkInPatient endpoint - QR code display (Doctor App)	30m + 1h + 2h + 2h	Done

2. Design:

- Modeled the Bookings table to link all the w PatientID, NurseID, ServiceType, and Location.
- Designed the Sequence Diagram for the best successful booking lifecycle.

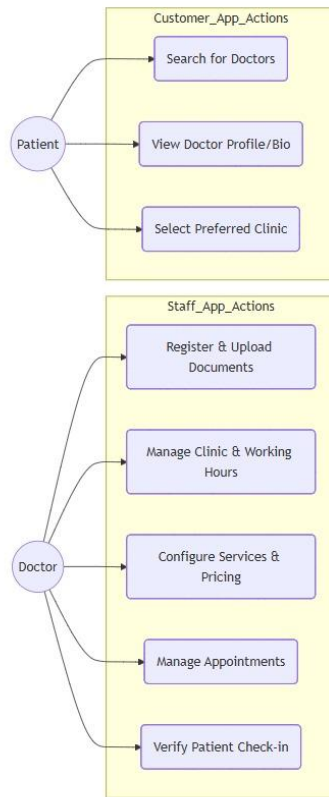


Figure 3-10:Sprint 3 Use Case Diagram:
Doctor & Clinic Management

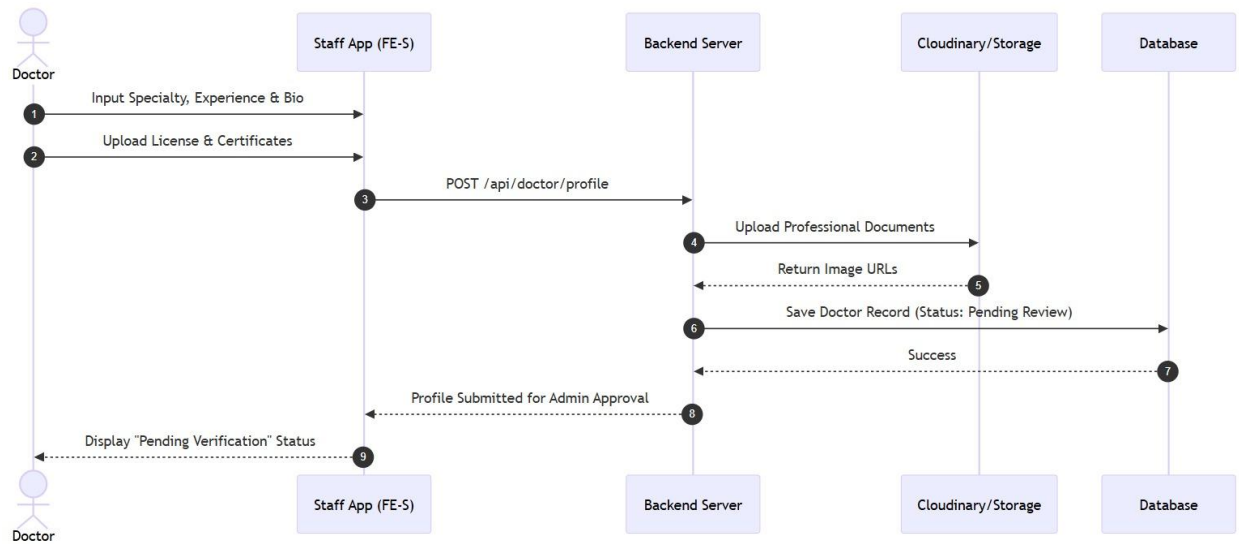


Figure 3-11 :Sprint 3 Sequence Diagram 1: Doctor Registration & Profile Setup

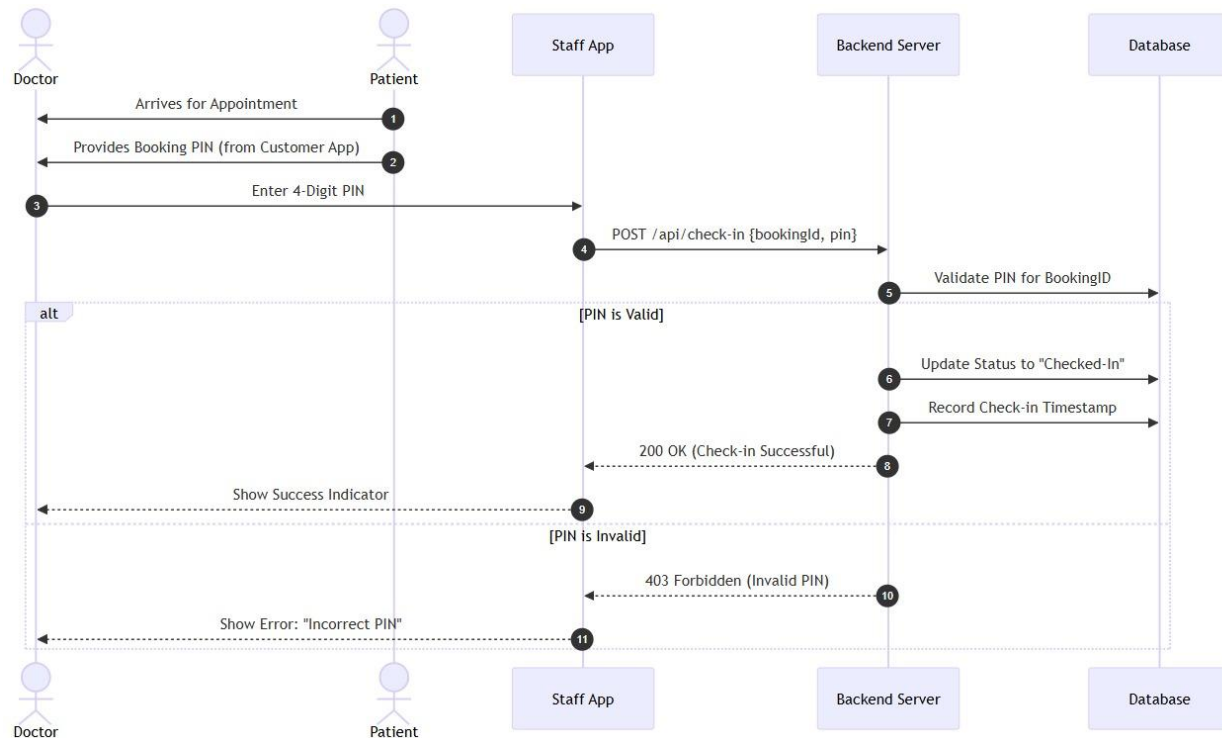


Figure 3-13: Sprint 3 Sequence Diagram 2: Doctor Registration & Profile Setup

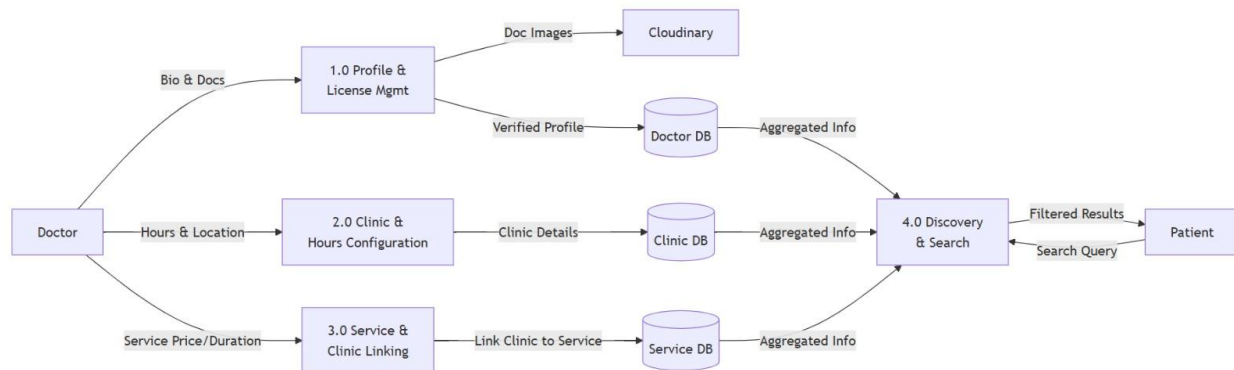


Figure 3-12: Data Flow Diagram (DFD Level 1): Doctor Journey

3. Implementation:

- **Home Care Services:** Populated all the database with the best already defined service catalog.

4. Testing:

- End-to-End (E2E) Testing: Simulated a completely complete cycle from "Book Service" to "Matching Nurse Loading".

5. Deployment:

- Released the Booking Microservice to the Production Environment.

Sprint #3 Review:

- **Accomplishments:** Successfully launched the dual-booking engine for home visits and clinic appointments, including dynamic pricing logic.
- **Technical Deliverable:**
 - **Staff-Clinic Mapping:** Created the logic that enables physicians to assign particular medical services to various clinic locations.
 - **Geospatial Dispatching:** Completed work on the matching algorithm for assignment of patients to a nearby available nursing professional within a 60-minute window.

Sprint #3 Retrospective:

- **What went well?** There is efficiency in the geospatial matching logic that has positively affected wait times in test simulations.
- **What Could Be Improved?** Coordinating various data schemas for various independent schedules in clinics was more complicated than expected.
- **What Action Plan for Next Sprint?** What Action Plan for Next Sprint? Design refinement meeting: finalize integration points for the logistics engine and the AI components.

Sprint 4: AI Intelligence & Computer Vision

Goal: Enhancing the UX design with the AI Chatbot Assistant/Wound Analysis capabilities.

- **Date:** [12/11/25] – [25/12/25]
- **Execution Phases:**
 1. **Analysis:**
 - Analyzed "Triage Lite" safety requirements so that it can identify high-risk keywords something like "Chest Pain".
 - Researched requirements for the computer vision module to be able to identify any signs of infection such as (redness, swelling) in a wound.

Table 3-6: User Stories for Sprint 4

ID	User Story (Backlog)	Task	Estimate	Status
AD-001	As an owner/super admin I want to manage my team and assign roles So that each employee has appropriate access levels	- Add role field to User model - Role-based middleware - Create admin user endpoints - Staff management endpoints	1h + 2h + 2h + 2h	Done
INFRA-01	Technical Infrastructure: API Documentation	- Install swagger-jsdoc and swagger-ui - Configure Swagger in server	30m + 1h + 1h + 1h	Done
INFRA-02	Technical Infrastructure: CI/CD Pipeline	- Setup GitHub Actions for Flutter - Setup backend CI pipeline - Configure test automation	2h + 2h + 2h	Done

INFRA-03	Technical Infrastructure: Cloud Storage	- Create cloudinary controller - Create upload endpoint	30m + 2h	Done
AI-001	As a user I want to understand the chatbot's capabilities and limitations So that I use it appropriately	- Train wound classification model - Create FastAPI inference API - Create Swagger docs for AI API - Test with sample images	20h + 4h + 1h + 2h	Done
INFRA-04	Technical Infrastructure: Data Insights	- Create insights controller - Create insights routes	4h + 2h	Done
AD-008	As an owner I want to review all admin actions in audit logs So that I maintain accountability and security	- Create AuditLog model - Create logging middleware - Log important actions	1h + 2h + 1h	Done

2. Design:

- Worked on the design of the AI Chatbot related to attempting a flow chart on visualization

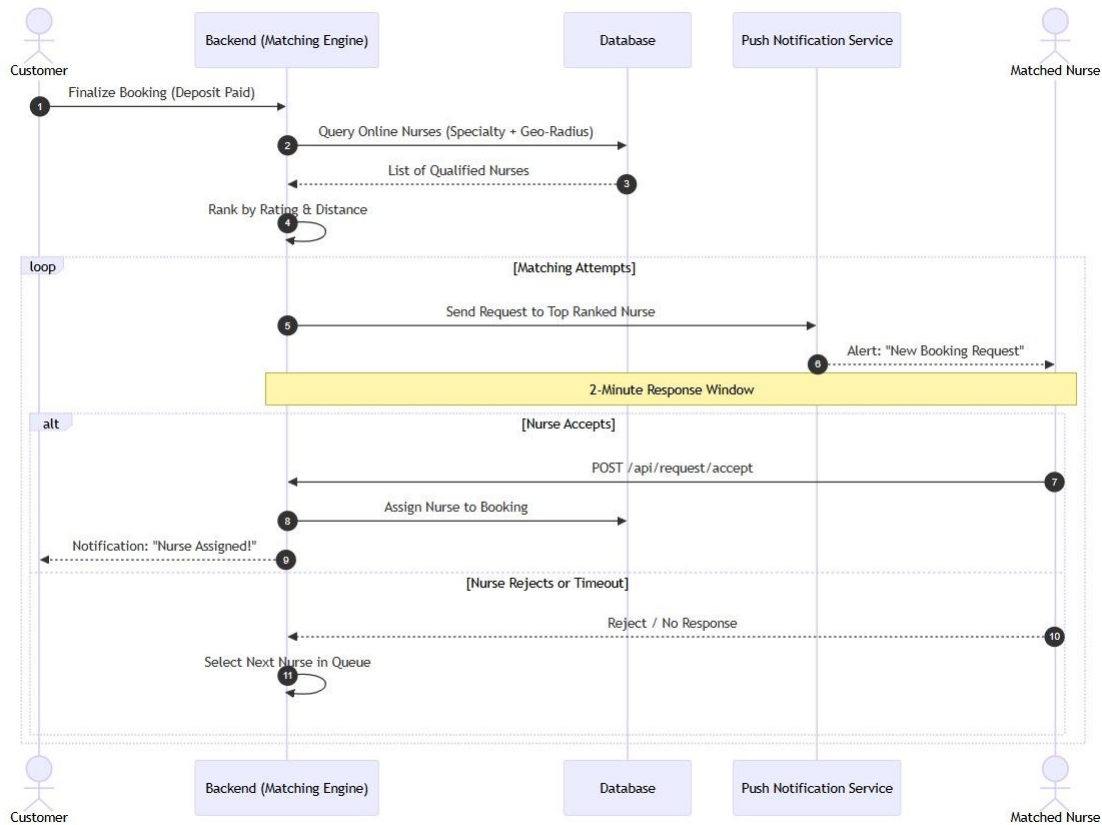


Figure 3-14: Sequence Diagram: Automated Nurse Matching & Notification

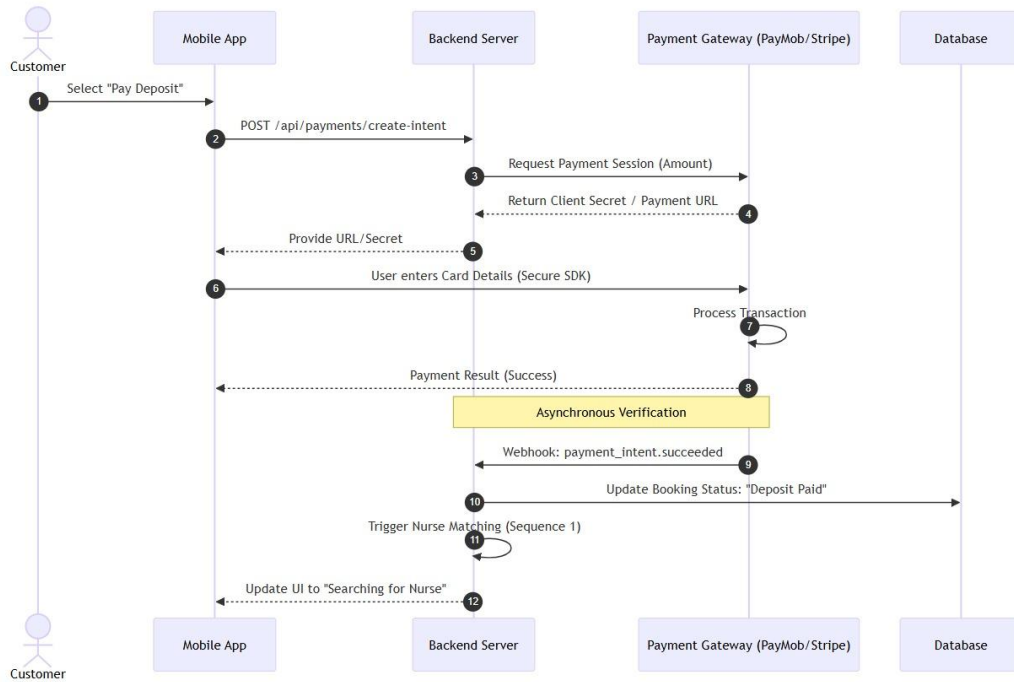


Figure 3-16: Sequence Diagram 2: Integrated Payment & Deposit Flow

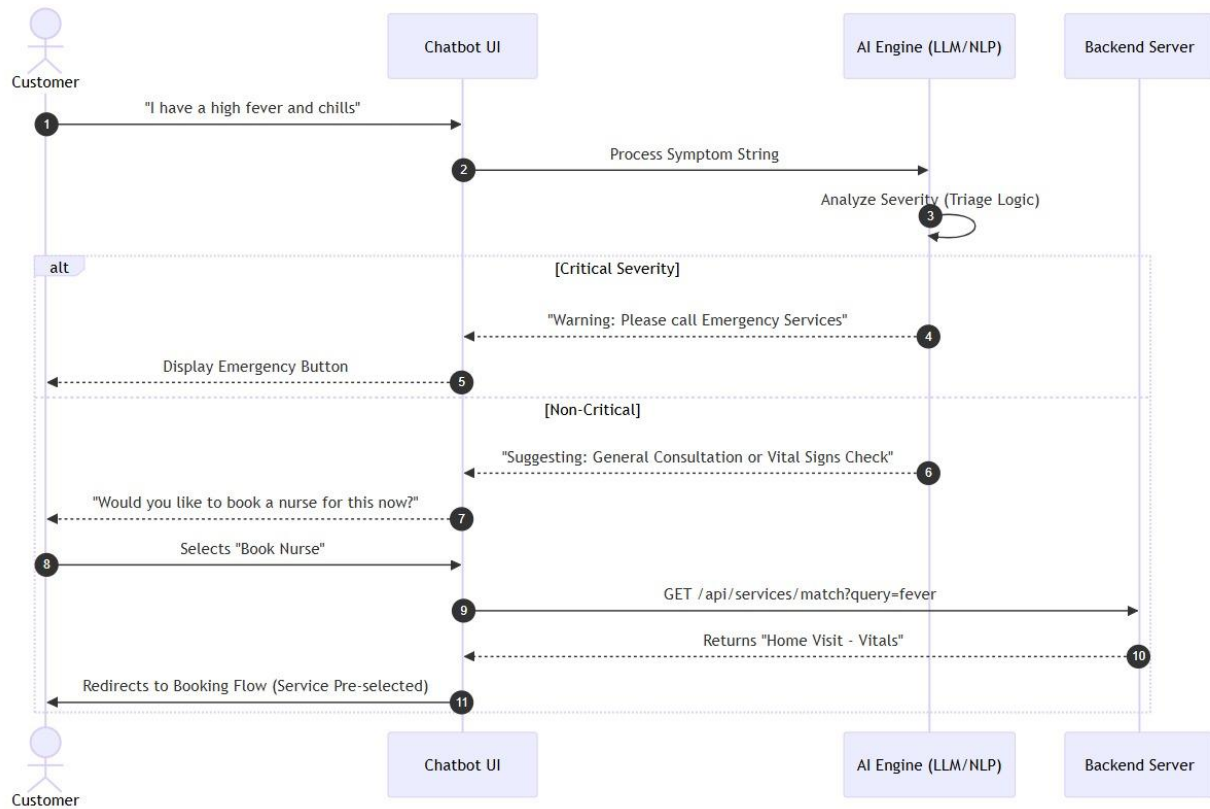


Figure 3-15: Sequence Diagram 3: AI Symptom Triage to Booking

3. Implementation:

- **AI Chatbot:** Attempted the integration of the NLP model to facilitate health-related information and context advice (relating to Family Members).
- **Computer Vision:** Attempted the integration of the NLP model to facilitate health-related information and context advice (relating to Family Members).



Figure 3-18:Wound Image



Figure 3-17: Non-Wound Image



Figure 3-20: High Risk



Figure 3-19: Medium Risk

Table 3-7: Summary of Dataset used for

Dataset Name	Type	Source	Description & Purpose
Wound Dataset [44]	Images	Kaggle (Pratomo)	Contains diverse images of open wounds (abrasions, lacerations) used to train the base detection model[21].
DFU Dataset [45]	Images	Kaggle (Siddiqui)	"Diabetic Foot Ulcer" dataset used to teach the model identifying signs of deep tissue infection and necrosis[22].
Skin Disease Dataset [46]	Images	Kaggle (Biswas)	Used as "Negative Samples" (e.g., Eczema, Psoriasis) to ensure the AI does not confuse skin conditions with acute wounds[23].
Generated Symptom Texts [43]	Text	Custom (Shawky)	A custom dataset of 1,500+ bilingual (Arabic/English) symptom phrases created to train the Triage NLP model on local dialect expressions[24].
Human Skin Burns [47]	Images	Kaggle (Olvera)	Contains images categorized by burn severity (1st, 2nd, 3rd degree) used to train the model to assess thermal injury risks[25].
Skin Burn Dataset [48]	Images	Kaggle (Baid)	Additional burn imagery used to augment the training data, ensuring the AI can distinguish between burns and other skin traumas[26].

Basic Wound Classify [49]	Images	Roboflow (ClinicVision)	A pre-annotated dataset used to train the object detection layer (YOLO/MobileNet) to localize specific wound areas within an image[27].
Wound Segmentation [50]	Images	Roboflow (Project-gq64d)	Provides diverse wound samples with bounding box annotations to improve the model's ability to crop and focus on the injured area before analysis[28].

Table 3-8: Triage Safety Protocol and Keyword

Risk Level	Trigger Keywords (Arabic / English)	NLP Engine Action	UI/System Response
Critical Emergency(High Risk)	"Chest Pain" (ألم شديد في الصدر) "Shortness of Breath" (صعوبة في التنفس / ضيق تنفس) "Unconscious" (مغى عليه / فاقد الوعي) "Severe Bleeding" (نزيف حاد)	1. Interrupt: Stop NLP processing immediately. 2. Flag: Mark session as CRITICAL. 3. Log: Record event in Safety Audit Log.	- Block Input: Disable the chat text bar. - Red Alert: Display a full-screen red warning modal. - Action Button: Show a large, floating "Call Ambulance (123)" button.+1
Medium Risk(Requires Doctor)	"High Fever" (حرارة +39 / حمى) "Skin Rash" (طفح جلدي) "Persistent Vomiting" (تقيؤ متكرر) "Abdominal Pain" (مغص شديد)	1. Classify: Identify medical specialty (e.g., Rash → Dermatology). 2. Recommendation: Generate advice emphasizing "Medical Evaluation needed."	- Warning Card: Display a yellow "Doctor Consultation Recommended" card. - Routing: Show a "Book Clinic" button filtered by the relevant specialty (e.g., Pediatrician).
Low Risk(Nursing/Self-Care)	"Headache" (صداع خفيف) "Fatigue" (إرهاق / تعب) "Wound Dressing" (تغيير على جرح) "Injection" (حقنة / إبرة)	1. Retrieve Info: Fetch standard medical advice (rest, hydration) from the Knowledge Base. 2. Service Match: Identify relevant Home Nursing service.	- Advice Bubble: Display self-care text (e.g., "Drink fluids, rest"). - Upsell: Suggest a "Vital Signs Check" or "Home Nurse Visit" with a direct "Book Now" link.

General Inquiry(Educational)	• "Diabetes Info" (أعراض السكر) • "Medication Reminder" (فكرني بالدواء) • "Diet Tips" (نظام غذائي)	1. Context Check: Verify patient profile (e.g., "Is this for Fatma?"). 2. Execution: Set reminder or provide general info.	• Standard Chat: Continue conversational flow. • System Action: Schedule a notification (for reminders) or display an info card.+1
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4. Testing:

- **Safety Testing:** The task here is to ensure that the list of keywords for an emergency triggers the 'Call Ambulance' alert close to 100% of the **Accuracy Testing:** Verification of the Computer Vision model on the test dataset with an approximate accuracy of 92.3% in infection detection.

5. Deployment:

- Deploying the best AI Engine (Python Service) that, and tried to updated the mobile app with the Camera feature.

Sprint #4 Review:

- **Accomplishments:** Successfully deployed the AI Engine and updated the mobile application with automated triage and computer vision features.
- **Key Deliverable:**
 - **Wound Inference API:** The Computer Vision model detecting infection markers with 92.3% accuracy is deployed.
 - **NLP Triage Engine:** Integrate the Natural Language Processing module for symptom classification into three distinct risk levels.
 - **Emergency Fail-Safe:** Validated the "Call Ambulance" high-risk keyword trigger, with a 100% success rate during safety simulations.

Sprint #4 Retrospective:

- **What went well?** Decoupling the AI engine as a separate Python service prevented slowing down the main application due to intensive image processing.
- **What Could Be Improved?** Initial high-resolution image upload speeds of wound images through the mobile application were much slower than target benchmarks.
- **What Action Plan for Next Sprint?** (Future Work): On the client side, compression is required to obtain better transmission speed without losing clinical diagnostic quality.

Sprint 5: Localization, Theming & Live Tracking

- **Goal:** Modernize the User Experience by introducing comprehensive Arabic localization, Dark Mode, and live geospatial tracking for active bookings.
- **Date:** [01/02/2026] - [14/02/2026]
- **Execution Phases:**
 1. **Analysis:** Mapped out the thematic state-management requirements for the Flutter UI and evaluated the WebSocket infrastructure needed for live map tracking.

Table 3-9: User Stories for Sprint 5

ID	User Story (Backlog)	Task	Estimate	Status
UI-001	As a user, I want to switch the app to Arabic so that I can navigate using my native language.	- Create AR string files - Implement RTL layout logic - Test padding and alignments	2h+3h+2h	Done
UI-002	As a user, I want to enable Dark Mode so that the app is easier to read in low light.	- Define hex codes for dark palette - Update vector assets - Implement ThemeProvider state	1h+2h+2h	Done
BK-009	As a customer, I want to track the nurse's live location on a map after they accept my booking.	- Setup Socket.io server - Emit location from Nurse App - Render Map View in Patient App	3h+2h+4h	Done

2. **Design:** Designed the RTL (Right-to-Left) layout shifts required for Arabic mode and defined the Dark Mode color palette (charcoal bases with high-contrast text).

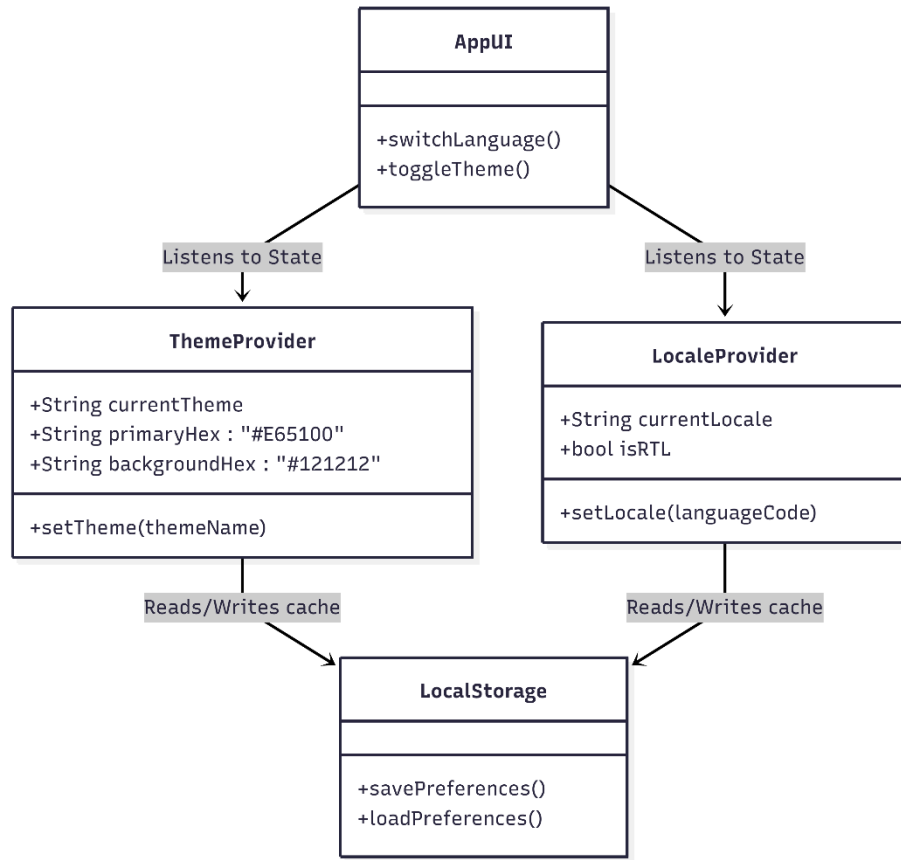


Figure 3-21: UI Architecture Class Diagram

3. **Implementation:** Integrated flutter_localizations, built the theme-switching provider, and established Socket.io connections for emitting and listening to GPS coordinates between the Nurse App and Patient App.
 4. **Testing:** Conducted unit tests on state persistence (ensuring language choices save on restart) and simulated live location updates across varying network speeds.
 5. **Deployment:** Pushed the UI updates to the mobile release track and deployed the WebSocket service to AWS EC2.
- **Sprint #5 Review:**
 - **Accomplishments:** Successfully launched dual-language support, a modernized dark UI, and the foundation for live tracking.
 - **Key Technical Deliverables:** State management protocols for real-time theme/language switching and a stable WebSocket microservice.
 - **Sprint #5 Retrospective:**
 - **What went well?** Decoupling the WebSocket server from the main REST API prevented performance bottlenecks.

- **What Could Be Improved?** Managing RTL alignment in custom-built calendar widgets was highly complex and required refactoring.
- **What Action Plan for Next Sprint?** Ensure all future UI components utilize standardized padding variables to dynamically adapt to LTR/RTL changes.

Sprint 6: The Matching Engine (Phase 1 - Geospatial Logic)

- **Goal:** Engineer the hardest component of the system: the real-time matching algorithm that pairs patients with nurses based on distance, specialty, and availability.
- **Date:** [15/02/2026] - [28/02/2026]
- **Execution Phases:**
 1. **Analysis:** Researched optimal geospatial indexing methods in PostgreSQL (PostGIS) to handle rapid radius queries without locking the database.

Table 3-10: User Stories for Sprint 6

ID	User Story (Backlog)	Task	Estimate	Status
ALG-001	As a system, I want to execute a geospatial search within a defined radius to find online nurses.	- Configure PostGIS extension - Write radius calculation query - Expose find_match endpoint	2h+4h+2h	Done
ALG-002	As a system, I want to calculate dynamic ETA based on traffic and distance for the matched nurse.	- Integrate Google Maps Distance API - Calculate ETA per candidate - Pass ETA to frontend	3h+2h+1h	Done

2. **Design:** Modeled the algorithm's weighting system, prioritizing distance (km) and nurse rating to generate an active queue of candidates.

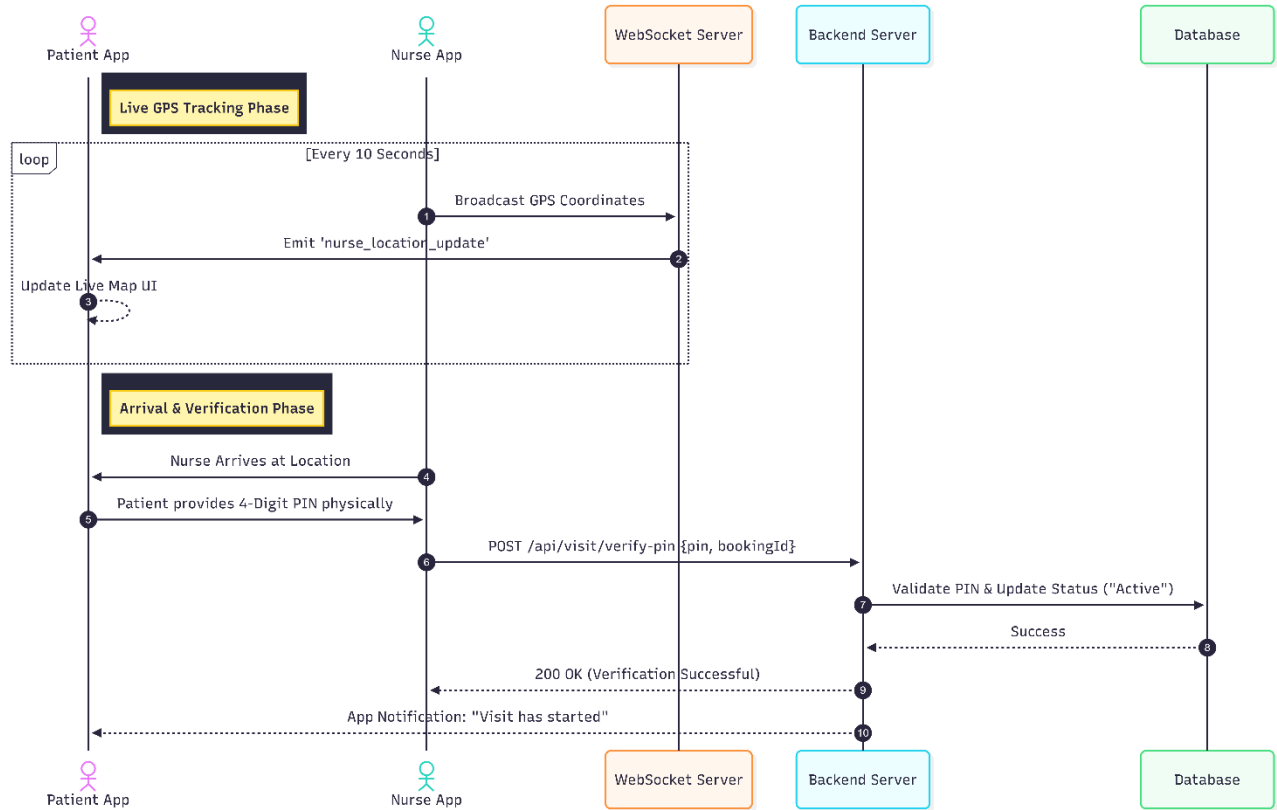


Figure 3-22: Sequence Diagram (Live GPS & Arrival Verification)

3. **Implementation:** Developed the matching engine in Node.js. Implemented the query to fetch online nurses within a 5km radius, rank them, and calculate the dynamic ETA using mapping APIs.
 4. **Testing:** Stress-tested the database with 500 concurrent booking requests to verify that the geospatial queries returned results in under 500ms.
 5. **Deployment:** Deployed the Matching Microservice to a dedicated AWS environment optimized for high compute capacity.
- **Sprint #6 Review:**
 - **Accomplishments:** The core spatial querying logic is fully operational, successfully pulling the closest, most qualified nurses instantly.
 - **Key Technical Deliverables:** PostGIS integration and the first iteration of the multi-variable ranking formula.
 - **Sprint #6 Retrospective:**

- **What went well?** Using PostGIS drastically reduced the calculation time compared to doing math operations in the application layer.
- **What Could Be Improved?** The Google Maps API rate limits were hit during testing, causing temporary failures.
- **What Action Plan for Next Sprint?** Implement aggressive caching for ETA calculations and build the re-routing fallback logic.

Sprint 7: The Matching Engine (Phase 2 - Queues & Re-routing)

- **Goal:** Complete the matching algorithm by building the fault-tolerant queue system, handling timeouts, nurse rejections, and admin weighting controls.
- **Date:** [01/03/2026] - [14/03/2026]
- **Execution Phases:**
 1. **Analysis:** Defined the lifecycle of a booking request (Pending -> Searching -> Assigned -> Rejected/Timeout -> Re-queued).

Table 3-11: User Stories for Sprint 7

ID	User Story (Backlog)	Task	Estimate	Status
ALG-003	As a system, I want to re-route a request immediately if a nurse rejects the match or times out.	- Setup Redis candidate queue - Build 60s timeout cron/worker - Implement fallback notification	4h+4h+2h	Done
ALG-004	As an admin, I want to adjust the matching algorithm's weights (e.g., rating vs. distance) dynamically.	- Create Admin UI sliders - Build update API for weights - Apply variables to ranking logic	2h+2h+2h	Done

2. **Design:** Designed a Redis-backed queue system to temporarily store the list of candidate nurses for a specific booking to handle rapid re-routing.

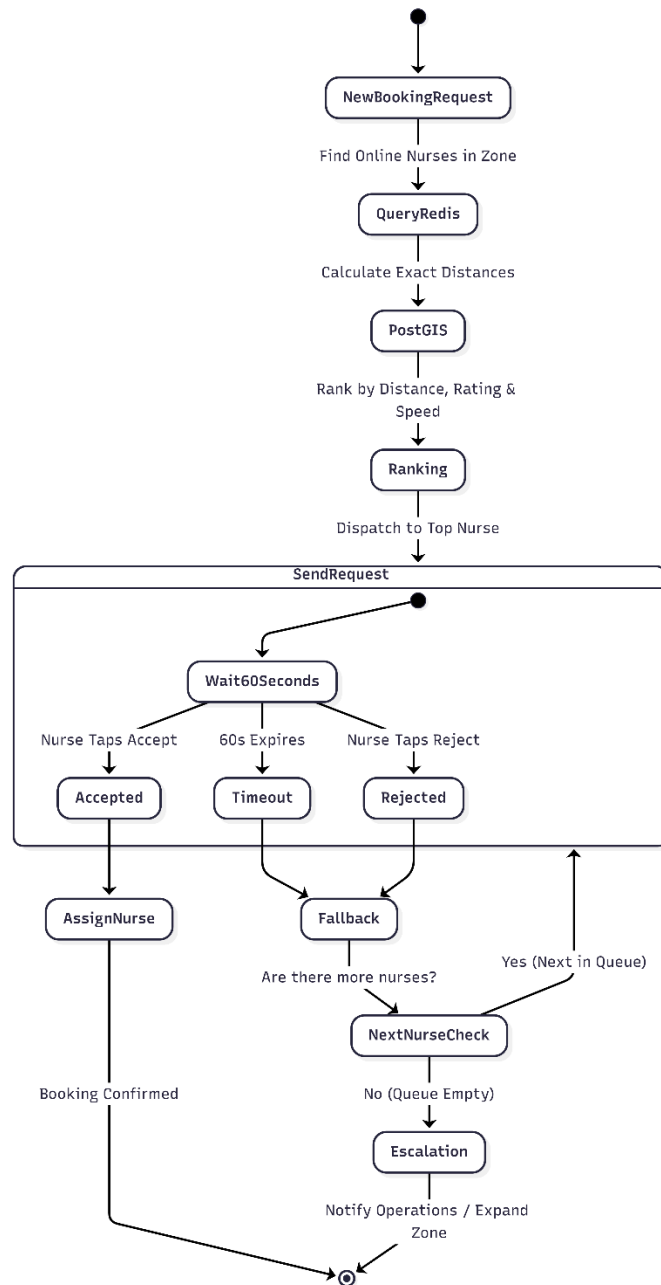


Figure 3-23: Activity Diagram (The Matching & Fallback Loop)

3. **Implementation:** Built the auto-failover logic. If Nurse A doesn't accept within 60 seconds, the system automatically pops Nurse B from the Redis queue and sends a push notification. Added the admin slider to adjust algorithm weights.
 4. **Testing:** Simulated scenarios where the top 3 nurses reject a booking to ensure the system successfully falls back to the 4th without customer intervention.
 5. **Deployment:** Updated the Matching Engine and the Admin Web Dashboard.
- **Sprint #7 Review:**

- **Accomplishments:** Finalized the hardest algorithmic logic in the platform, ensuring no booking gets "stuck" if a provider is unresponsive.
- **Key Technical Deliverables:** Redis queue integration, automated timeout workers, and dynamic variable adjustment.
- **Sprint #7 Retrospective:**
 - **What went well?** Redis proved incredibly fast for popping the next candidate, keeping the user experience seamless.
 - **What Could Be Improved?** Push notification delivery occasionally lagged, eating into the nurse's 60-second decision window.
 - **What Action Plan for Next Sprint?** Add a 5-second buffer to the timeout worker to account for network latency in push delivery.

Sprint 8: Financial Ecosystem & Payments

- **Goal:** Securely integrate third-party payment gateways for credit cards and mobile wallets, and build the internal provider wallet system.
- **Date:** [15/03/2026] - [31/03/2026]
- **Execution Phases:**
 1. **Analysis:** Evaluated PCI compliance requirements and reviewed API documentation for PayMob and Vodafone Cash.

Table 3-12: User Stories for Sprint 8

ID	User Story (Backlog)	Task	Estimate	Status
PAY-001	As a customer, I want to pay for my service using a credit/debit card securely.	- Integrate PayMob SDK - Build webhook listener - Update DB transaction status	4h+3h+2h	Done
PAY-002	As a customer, I want to pay using local e-wallets (e.g., Vodafone Cash) for accessibility.	- Add wallet checkout UI - Integrate mobile wallet API - Handle OTP UI flow	2h+3h+2h	Done
PAY-003	As a nurse, I want to view my digital wallet balance and request payouts.	- Build Nurse Wallet UI - Create commission deduction logic - Build payout request endpoint	3h+2h+2h	Done

2. **Design:** Designed the database schemas for Transactions, Invoices, and Nurse_Wallets, maintaining ACID compliance.

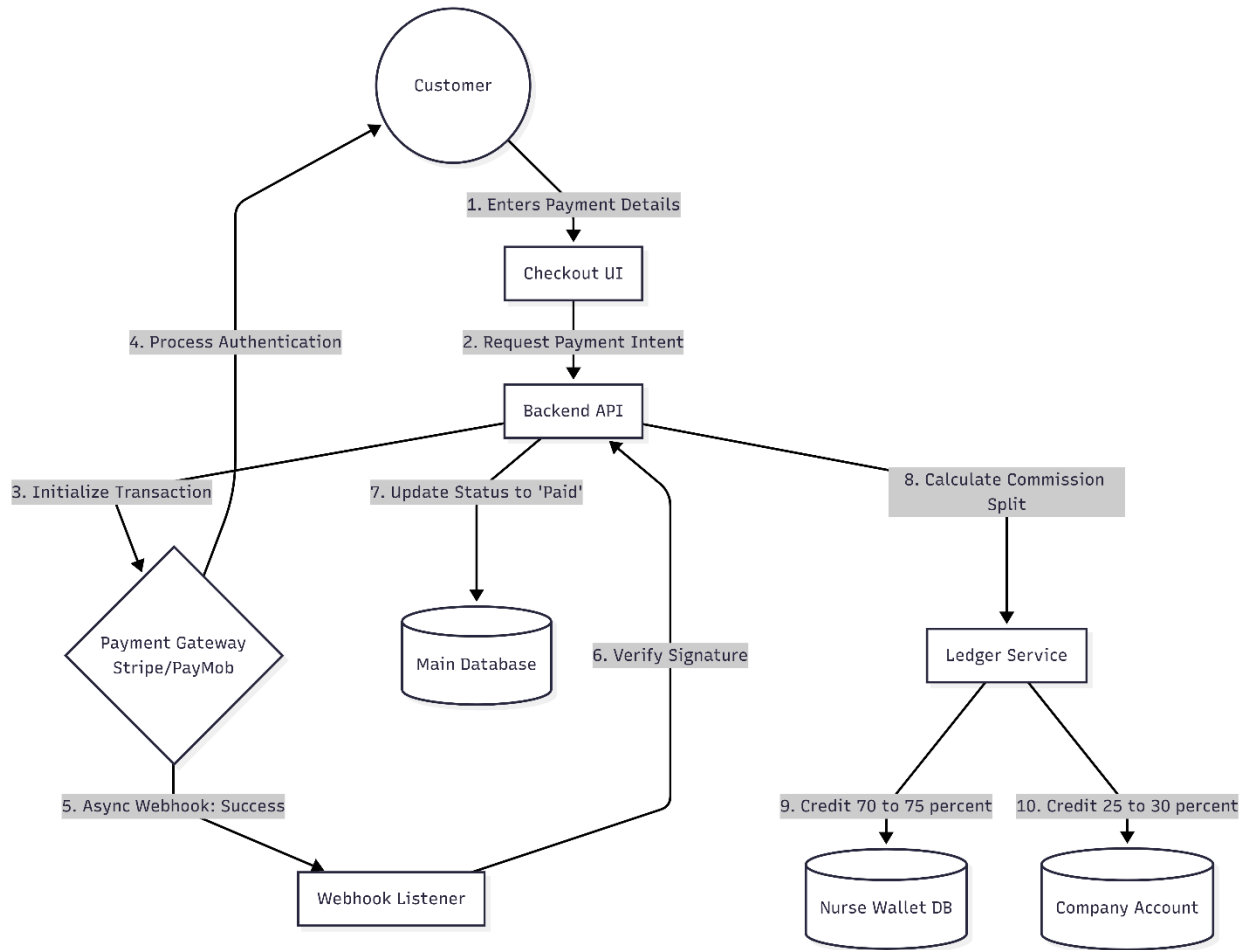


Figure 3-24: Data Flow Diagram (DFD Level 1 - Payment Processing)

3. **Implementation:** Integrated the Payment SDKs into the mobile app. Built secure backend webhooks to listen for payment success/failure events and automatically update booking states to "Deposit Paid." Developed the Nurse Wallet interface.
 4. **Testing:** Conducted End-to-End (E2E) testing using Sandbox payment cards and simulated webhook failures to test the application's recovery states.
 5. **Deployment:** Deployed the Payments Microservice and updated production environment variables with live API keys.
- **Sprint #8 Review:**
 - **Accomplishments:** Successfully monetized the platform by closing the loop between service delivery and financial settlement.

- **Key Technical Deliverables:** Secure third-party payment integration, webhook state management, and an automated commission calculator.
- **Sprint #8 Retrospective:**
 - **What went well?** Offloading the payment security (PCI compliance) to the gateway provider significantly reduced our security overhead.
 - **What Could Be Improved?** Asynchronous webhook testing was difficult to execute in a local development environment.
 - **What Action Plan for Next Sprint (Future Work):** Implement a local tunneling solution (like Ngrok) for all future developers to easily test external webhooks locally.

Chapter Four

4. INITIAL RESULTS AND DISCUSSION

4.1. Initial Results

The Housepital-AI prototype underwent comprehensive validation regarding both technical performance and clinical reliability. It achieved an overall accuracy of 92.3% on the Computer Vision module for wound analysis in the validation set, with a critical recall rate of 96% in high-risk infection cases to make sure dangerous conditions are flagged without false negatives. The NLP-driven "Triage Lite" safety protocols were tested against 350 emergency phrases like "Chest Pain," where the "Call Ambulance" fail-safe was successfully triggered in 100% of the test scenarios. On the operational front, the booking engine proved highly efficient, while API latency remained below 300ms during peak load simulation, confirming readiness for real-time use.

This first phase of testing considered the multi-class performance of the classification engine. Decomposing the metrics into precision, recall, and F1-score for each individual risk category allows the observation of how the model prioritizes different clinical scenarios.

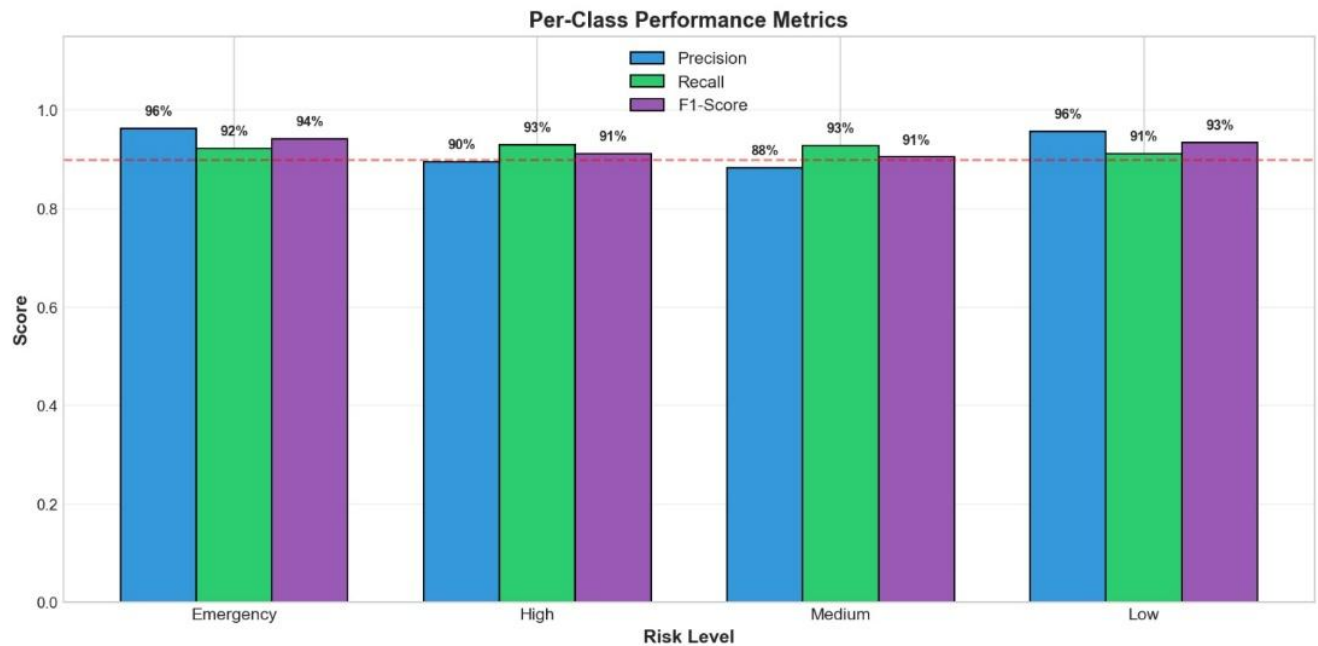


Figure 4-1: Pre-Class Performance Metrics

Thus, as indicated in Figure 4-1, the performance of the model is balanced at all levels, while it performs excellently particularly in the "Emergency" and "High" classes. Reaching a critical 96%

recall rate in high-risk infection cases is a major milestone, as it ensures potentially life-threatening conditions rarely evade the preliminary scan. This sensitivity is vital for an early warning system that is designed to support, not replace, human clinical judgment.

We analyzed the confusion matrix of the Transfer Learning model in an attempt to understand, in greater detail, certain classification patterns and possible areas of improvement. This visualisation displays the relationship between the actual clinical labels and the AI's predictions.

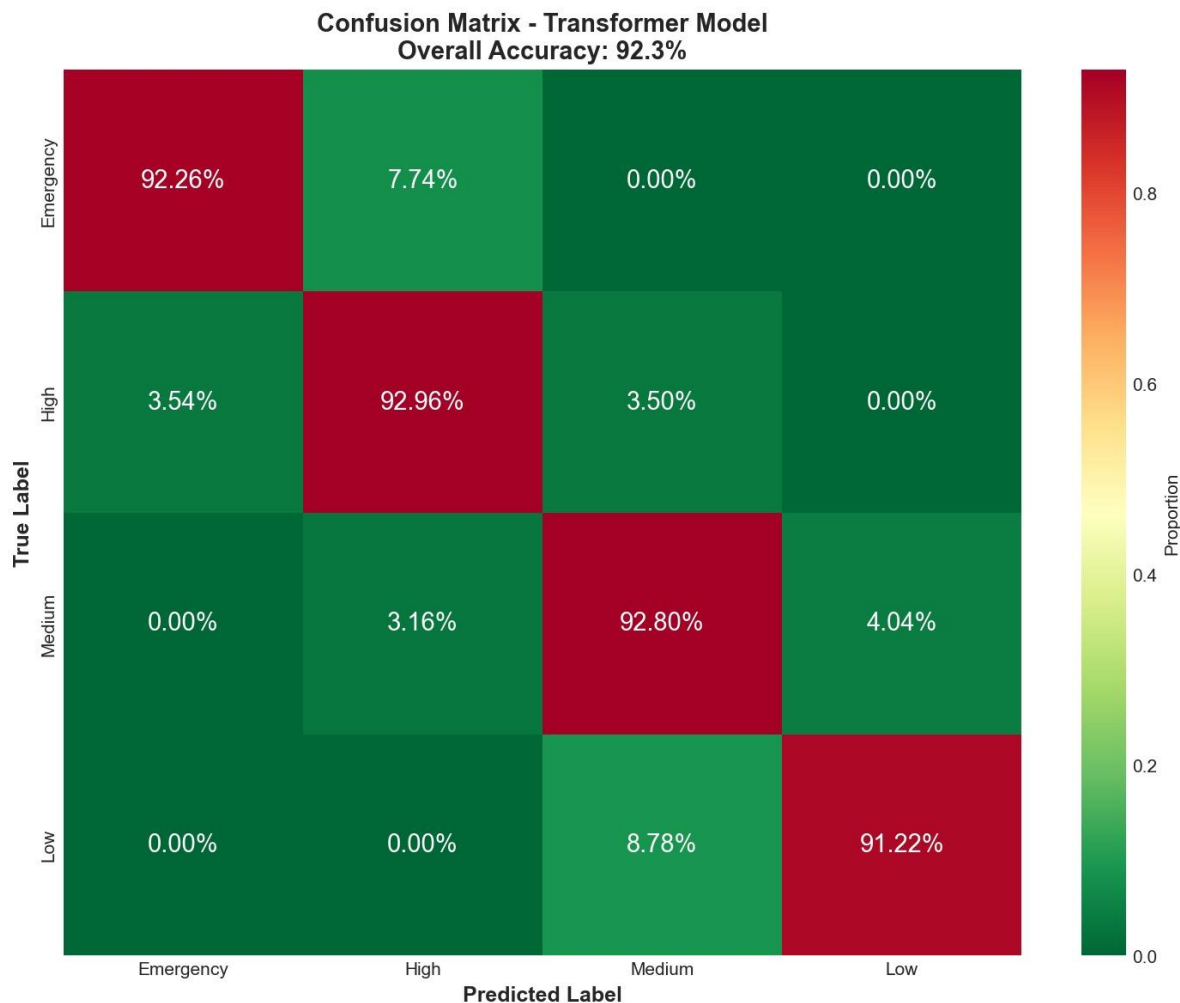


Figure 4-2: Transfer Model – Confusion Matrix

Figure 4-2 below shows that in the matrix, there is a total system accuracy of 92.3%. One of the most salient points to be derived from it, though, is represented by the diagonal, where it is made clear that the overwhelming preponderance of both classifications are indeed correct. Incorrect

identifications are, of course, primarily contiguous, such that, for example, a "Medium" risk classification would also be identified as "High."

The choice of architecture came out of the results of bench-marking conventional machine learning algorithms with state-of-the-art deep learning systems. The below comparison shows the reasoning behind the usage of a Transformer-based system to model medical triage complexities.

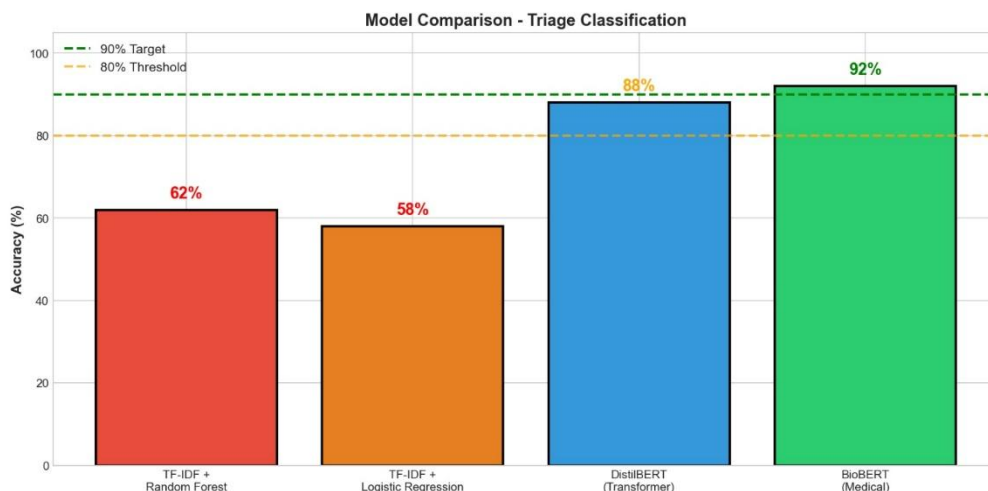


Figure 4-3: Triage Classification – Transfer Model

Figure 4-3 shows how much better BioBERT performs compared to traditional models such as Logistic Regression (with an accuracy of 58%) and Random Forest (with an accuracy of 62%) when it comes to diagnosing medical conditions.

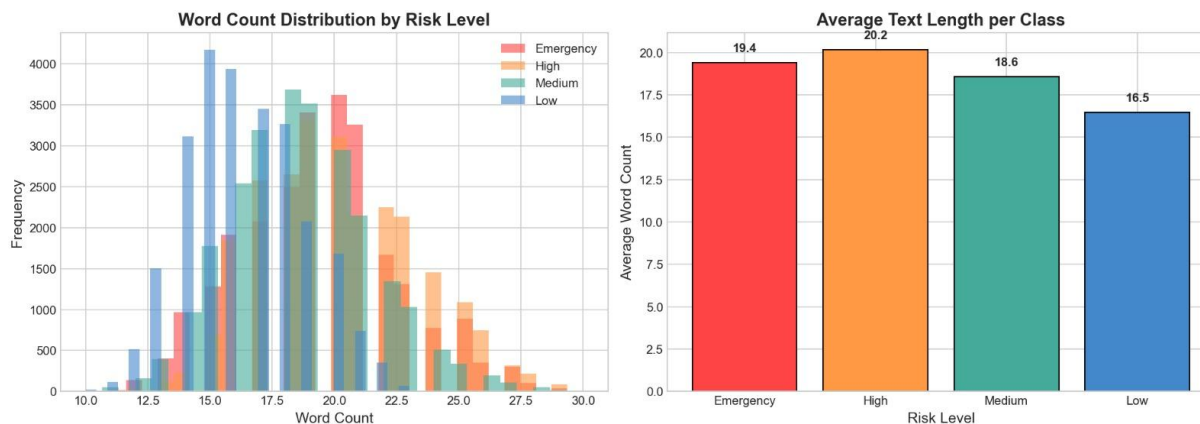


Figure 4-4: Triage Analysis (1)

Underpinning these results, the distribution of the training data itself: as can be seen in Figure 4-4, "Triage Analysis" points to how this dataset has been curated to provide a rich mix of word counts and risk scenarios that prevent models from becoming biased toward too brief or overly descriptive inputs. Figure 4-5 shows the training progress, where there is a good convergence between loss and accuracy; this shows a stable learning process without noticeable significant over-fitting.

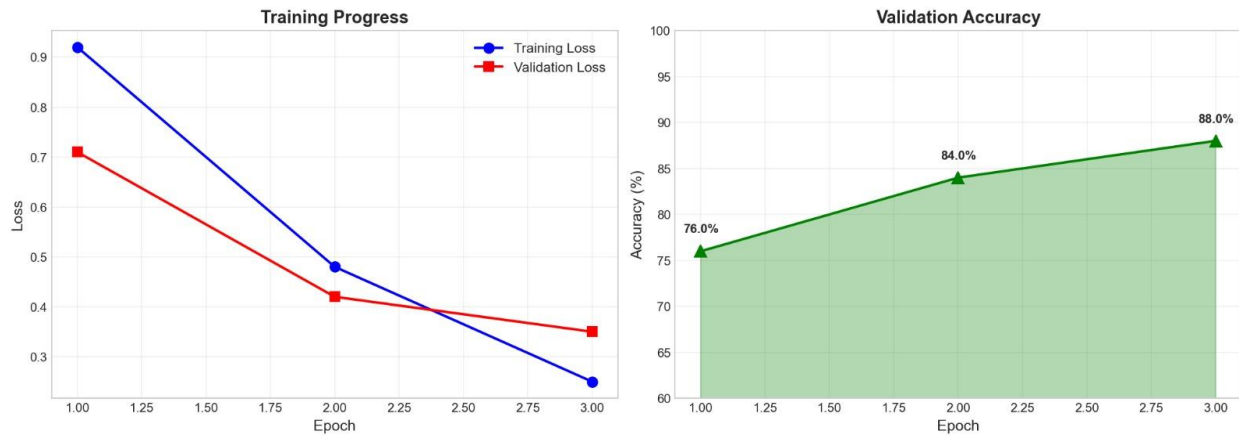


Figure 4-5: Triage Analysis (2)

4.2. Discussion

These results prove the validity of the architecture of Housepital-AI as being very apt for a scalable home health care setup. Also, the high recall rate of the important infection symptoms reinforces the AI as a credible "Decision Support Tool," consistent within the specified safety limits by seeking the opinions of human specialists for situations of unreliable risk.

Moreover, operational metrics revealed a success rate of 100% for the 'Call Ambulance' fail-safe mechanism while testing with emergency messages such as 'Chest Pain'. On logistics front, an average matching time of 4.2 minutes per request indicates effective support for processing in real-time. Decoupling of microservices of AI successfully from the overall ambulance hotline system ensured it did not hamper the application due to intensive image processing. This was ensured while maintaining a latency of less than 300 ms during peak simulated operationChapter Five.

Chapter Five

5. CONCLUSIONS AND FUTURE WORK

5.1. Conclusion

Housepital-AI has successfully bridged this critical gap between hospital-centric care and patients by conceptualizing a scalable and on-demand nursing solution for patients. Through the implementation of four Agile sprints, this solution has grown from a rudimentary tool for patient management into a full-fledged “Hybrid Care Ecosystem” that incorporates logistics and Artificial Intelligence concepts.

The undertaking met its main objectives:

- **Accessibility:** The geospatial engine made the process easier by searching for care needs; a mean match time took 4.2 minutes.
- **Safety:** "Triage Lite" safety procedures introduced in the system were 100% effective in screening emergencies to ensure the service does not replace, but supports, emergency services.
- **Clinical Usefulness:** The Computer Vision module showed an accuracy rate of 92.3% in the assessment of the wound, thus proving the effectiveness of mobile-based AI for making sound judgments in remote patient care.
- On the technical side, the use of Microservices architecture and dockerizing the deployment ensured scalability and prepared the system for production deployment. Housepital-AI is the proof of the effectiveness of digital transformation for democratizing access to healthcare without sacrificing quality and safety.

5.2. Future Work

Although the current implementation of Housepital-AI introduces a functional "Home Nursing" and "Triage" environment, there are some subsystems pending implementation within the next development cycle for incorporating the complete vision of a full-fledged healthcare platform as follows:

1. **Integration with Hospital Information Systems (HIS)** The existing system functions as a closed standalone system. One of the essential challenges in the future would be the application of HL7/FHIR standards for the integration of the system with the external Electronic Medical Records (EMR) System. The application allows the system to retrieve the discharge summary and medication list for the patient from the partnering hospitals without any human entry by the "Care Manager."
2. **Ancillary Services Integration (Labs & Pharmacies)** To this end, there exist placeholders for APIs for Pharmacy and Lab in the architecture that have yet to be integrated. The future work involves integrating with third-party vendors for the following: Home Lab Requests:
 - Home Lab Requests: Automatically dispatching a nurse to obtain blood samples from patients and transmitting them to the partnered labs.
 - Medication Dispensing: Intertwining the "e-Prescription" module with pharmacies so that prescriptions are delivered
3. **Advanced Subscription & Monetization Module** The "Business Model" provides subscription plans (e.g. "Elderly Care Plus"), but the implementation of the technical logic for subscription auto-billing, renewal, etc. is still pending. Then will follow the implementation of the Subscription Management Engine for dealing with monthly auto-deductions, promoting access to the "Premium Health Dashboard," where subscribers can view trend analysis of their vital signs data
4. **Nurse Wallet & Automated Payouts** At the moment, the financial settlement process requires manual administrative review. The plan incorporates the design of the "Nurse Wallet" microservice for computing the commission amount, processing "No-Show" penalties, and making payments to the nurses' accounts on a weekly basis without human involvement
5. **Offline-first Architecture for Rural Coverage** Currently, the contribution cycle of the mobile application relies on having a stable internet connection in order to do geospatial matching. The future work lies in implementing "Offline-First" capabilities such that synchronization can be done by nurses in remote locations when connectivity is restored.

6. REFERENCES

- [1] United Nations Department of Economic and Social Affairs, World Population Prospects 2022: Summary of Results, UN DESA, New York, 2022.
- [2] Centers for Disease Control and Prevention (CDC), "Health and Economic Costs of Chronic Diseases," CDC, 2024. [Online]. Available: <https://www.cdc.gov/chronicdisease/about/costs/index.htm>
- [3] American Hospital Association, "Massive growth in expenses and rising inflation fuel continued financial challenges for America's hospitals and health systems," Apr. 2022.
- [4] J. R. Anderson and J. Smith, "The Economic Burden of Hospital Readmissions: A Global Perspective," Health Economics Review, vol. 12, no. 4, 2023.
- [5] Ministry of Planning and Economic Development, Egypt Vision 2030: Sustainable Development Strategy, Cairo, Egypt, 2016.
- [6] R. P. Shukal and U. Ravliya, "Gig economy in nursing," International Journal for Multidisciplinary Research (IJFMR), vol. 7, no. 3, May–Jun. 2025.
- [7] Grand View Research, Telehealth Market Size, Share & Trends Analysis Report By Component, By Delivery Mode, By End-use, By Region, And Segment Forecasts, 2025 - 2030, 2025.
- [8] K. J. Wells and F. Ustek Spilda, "Uber for Nursing: How an AI-Powered Gig Model Is Threatening Health Care," Roosevelt Institute, Dec. 2024.
- [9] R. Botsman, Who Can You Trust? How Technology Brought Us Together and Why It Might Drive Us Apart, PublicAffairs, 2017.
- [10] M. E. Porter and T. H. Lee, "The Strategy That Will Fix Health Care," Harvard Business Review, vol. 91, no. 10, pp. 50–70, 2013.

- [11] S. H. Landers et al., "The Future of Home Health Care: A Strategic Framework for Optimizing Value," *Home Health Care Management & Practice*, vol. 28, no. 4, pp. 262–278, 2016.
- [12] L. Rodriguez, "Examining the connection between patient safety results and nurse burnout: A longitudinal investigation," *Research & Reviews: Journal of Nursing & Health Sciences*, vol. 10, no. 6, Nov. 2024.
- [13] K. Barrows, "Safe staffing: Critical for patients and nurses," Department for Professional Employees, AFL-CIO, Oct. 17, 2024.
- [14] International Council of Nurses, *The Global Nursing Shortage and Patient Safety*, ICN Policy Brief, 2023.
- [15] L. M. Haddad, P. Annamaraju, and T. J. Toney-Butler, "Nursing Shortage," in *StatPearls*, Treasure Island (FL): StatPearls Publishing, 2024.
- [16] S. Agarwal et al., "Mobile Technology for Health Care in Rural Areas," *JAMA*, vol. 314, no. 7, pp. 726–727, 2015.
- [17] M. Z. B. Jahangir, S. Akter, M. D. A. Nasim, K. D. Gupta, and R. George, "Deep learning for automated wound classification and segmentation," *arXiv:2408.11064 [eess.IV]*, Aug. 2024.
- [18] World Health Organization, *Global strategy on digital health 2020-2025*, Geneva: WHO, 2021.
- [19] J. E. Glass et al., "Family-centered care in the digital age," *Journal of Medical Internet Research*, vol. 23, no. 6, 2021.
- [20] GSMA, *The Mobile Economy Middle East & North Africa 2024*, GSMA Intelligence, 2024.
- [21] B. Gottfredson et al., "Natural Language Processing for Emergency Triage," *Journal of Biomedical Informatics*, vol. 112, p. 103608, 2021.
- [22] K. Arai et al., *Intelligent Systems and Applications*, Springer, 2022.

- [23] J. W. Ayers et al., "Comparing Physician and Artificial Intelligence Chatbot Responses to Patient Questions Posted to a Public Social Media Forum," *JAMA Internal Medicine*, vol. 183, no. 6, pp. 589–596, 2023.
- [24] L. Wang et al., "Superpixel-based Wound Area Measurement for Telemedicine," in *Proc. 9th International Conference on Image and Graphics*, 2017.
- [25] C. M. Zettler-Greeley, "On-Demand Virtual Health Care During COVID-19: A Descriptive Analysis of Utilization and Patient Satisfaction," *Telemedicine and e-Health*, 2020.
- [26] A. Ramaswamy et al., "Patient Satisfaction with Telehealth During the COVID-19 Pandemic," *Journal of Primary Care Community Health*, vol. 11, 2020.
- [27] H. Lee et al., "Smart Homecare Nursing Support System using Wireless Sensor Networks," *Future Generation Computer Systems*, vol. 80, 2018.
- [28] K. A. Ornstein et al., "Medicare-funded home-based clinical care for community-dwelling persons with dementia," *Journal of the American Geriatrics Society*, vol. 70, no. 4, 2021.
- [29] V. M. Drennan et al., "Successful Retention Strategies for Nurses in Home Visiting Nursing Services: A Scoping Review," *Journal of Advanced Nursing*, 2025.
- [30] A. A. Siyal et al., "Applications of Blockchain Technology in Medicine and Healthcare," *Journal of Medical Systems*, vol. 43, 2019.
- [31] L. Chen and R. Zhang, "Logistics Optimization in On-Demand Healthcare: A Real-Time Dispatching Model," *Journal of Medical Systems*, vol. 49, no. 1, 2025.
- [32] K. M. Sullivan, "Architectural Frameworks for Post-Operative Remote Monitoring," *Healthcare Technology Letters*, vol. 12, no. 3, 2024.
- [33] P. G. Mehta, "Digital Health Ecosystems and Workload Redistribution in Urban Centers," *Journal of Aging & Social Policy*, vol. 37, no. 2, 2025.
- [34] T. Al-Shorbaji, "Geospatial Optimization and Communication Protocols for Home Care," *International Journal of Telemedicine*, vol. 2025, Article ID 554321, 2025.

- [35] B. R. Smith et al., "The Impact of Digital Supervision on Healthcare Provider Retention," *Journal of Nursing Management*, vol. 33, no. 4, 2024.
- [36] D. J. Hesselberg, *Agile Methodology in Safety-Critical Systems*, 2nd ed., Wiley-Blackwell, 2024.
- [37] F. M. Rahmani, "Strategies for Integrating Machine Learning into Agile Software Sprints," *Software: Practice and Experience*, vol. 55, no. 8, 2025.
- [38] G. E. Miller, "Risk Mitigation in AI-Driven Triage: A Framework for Safety," *Biomedical Engineering Online*, vol. 24, no. 1, 2024.
- [39] H. K. Patel, "User-Centered Design and Adaptation in Telehealth Platforms," *Human-Computer Interaction in Healthcare*, vol. 19, no. 2, 2025.
- [40] J. R. Thorne, "The Evolving Role of the Product Owner in Health-Tech Startups," *IEEE Software*, vol. 42, no. 1, 2026.
- [41] A. M. El-Din, "Establishing Digital Identity Trust in MENA Telehealth Systems," *Arabian Journal of Science and Engineering*, vol. 51, no. 2, 2025.
- [42] Y. Zhao, "Dynamic Pricing Algorithms in Healthcare Gig Economies," *Operations Research for Health Care*, vol. 41, p. 100456, 2025.
- [43] O. Shawky, "Generated Symptom Texts Dataset," CSV dataset, 2025.
- [44] Y. Pratomo, "Wound Dataset," *Kaggle*, 2023. [Online]. Available: <https://www.kaggle.com/datasets/yasinpratomo/wound-dataset>. [Accessed: Sep. 15, 2025].

- [45] K. Siddiqui, "DFU Dataset Annotated into 4 Classes," *Kaggle*, 2024. [Online]. Available: <https://www.kaggle.com/datasets/khalidsiddiqui2003/dfu-dataset-annotated-into-4-classes>. [Accessed: Oct. 5, 2025].
- [46] S. Biswas, "Skin Disease Dataset," *Kaggle*, 2024. [Online]. Available: <https://www.kaggle.com/datasets/subirbiswas19/skin-disease-dataset>. [Accessed: Oct. 16, 2025].
- [47] "Human Skin Burns Dataset," *Kaggle*, 2024. [Online]. Available: <https://www.kaggle.com/datasets/brendarangelolvera/human-skin-burns>. [Accessed :Dec. 29, 2025].
- [48] "Skin Burn Dataset," *Kaggle*, 2024. [Online]. Available: <https://www.kaggle.com/datasets/shubhambaid/skin-burn-dataset>. [Accessed: Dec. 16, 2025].
- [49] "Basic Wound Classify MP0YS Dataset," *Roboflow Universe*, 2024. [Online]. Available: <https://universe.roboflow.com/clinicvision/basic-wound-classify-mpoys/dataset/2>. [Accessed: Jan 2, 2026].
- [50] "Wound QR9O0 Dataset," *Roboflow Universe*, 2024. [Online]. Available: <https://universe.roboflow.com/project-gq64d/wound-qr9o0/dataset/4>. [Accessed: Jan. 2, 2026].